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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 4

FURTHER INTEGRATION

SUGGESTED TIME: 6 PERIODS

IN THIS CHAPTER

This chapter develops integration as the inverse process of differentiation, from primitives and definite integrals to the standard techniques of the GCE Advanced Level: integration of rational functions, substitution, symmetry arguments and reduction formulae. It equips students with the computational toolkit required for area, volume and applied problems in the examination.

- By the end of this chapter I can define a primitive of a function and write the indefinite integral with its constant of integration.
- By the end of this chapter I can evaluate definite integrals using the Fundamental Theorem of Calculus and interpret them as areas.
- By the end of this chapter I can integrate rational functions using algebraic division and partial fractions.
- By the end of this chapter I can choose and carry out a suitable substitution to transform an integral into a standard form.
- By the end of this chapter I can exploit the symmetry of odd and even functions to simplify definite integrals over symmetric intervals.
- By the end of this chapter I can establish and use simple reduction formulae to compute integrals of powers of standard functions.

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THIS SIGNED COURSE CONTAINS

COURSE
complete, illustrated, step by step

METHODS
and worked examples

EXERCISES
exam style + solutions

SUMMARY SHEET
the essentials to remember

CONTENTS

1. Primitives and the Indefinite Integral
2. Definite Integrals and the Fundamental Theorem
3. Integration of Rational Functions
4. Integration by Substitution and Symmetry
5. Reduction Formulae and Synthesis
6. Integration activity
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CHAPTER OBJECTIVES

- By the end of this chapter I can define a primitive of a function and write the indefinite integral with its constant of integration.
- By the end of this chapter I can evaluate definite integrals using the Fundamental Theorem of Calculus and interpret them as areas.
- By the end of this chapter I can integrate rational functions using algebraic division and partial fractions.
- By the end of this chapter I can choose and carry out a suitable substitution to transform an integral into a standard form.
- By the end of this chapter I can exploit the symmetry of odd and even functions to simplify definite integrals over symmetric intervals.
- By the end of this chapter I can establish and use simple reduction formulae to compute integrals of powers of standard functions.
- By the end of this chapter I can verify an integral by differentiating the result.
- By the end of this chapter I can select the most efficient integration technique for a given GCE-style problem.

PREREQUISITES

- Differentiation of polynomials, trigonometric, exponential and logarithmic functions (Lower Sixth Pure Mathematics).

- The chain rule, product rule and quotient rule for derivatives.
- Standard derivatives table and algebraic manipulation of fractions and indices.
- Partial fractions decomposition (Lower Sixth algebra).
- Basic trigonometric identities, including double-angle formulae.

1 PRIMITIVES AND THE INDEFINITE INTEGRAL

An engineer at the SONARA refinery in Limbe must estimate the quantity of crude oil flowing through a pipe whose flow rate varies with time, while a farmer in Bafoussam wants to know the area of an irregular plot bounded by a curved river bank. In both cases, the quantity sought is an accumulation: knowing a rate of change, how do we recover the total? Integration answers this question, and this chapter turns the reverse of differentiation into a powerful computational machine. From rational fractions arising in electrical circuits to clever substitutions that tame complicated expressions, you will learn the techniques that make the GCE Advanced Level integrals collapse in a few lines.

The first step is to understand the simplest possible reversal of differentiation: given a function f, can we find a function F whose derivative is exactly f? Such a function is called a **primitive** (or antiderivative) of f, and the whole of this section is devoted to it.

1.1 The notion of a primitive

Intuition. Suppose a car travels along the Bamenda–Bafoussam road and its speed at time t is $v(t)$. The position of the car, $s(t)$, satisfies $s'(t) = v(t)$: position is a quantity whose rate of change is the speed. If we know the speed at every instant, can we reconstruct the position? Almost: we also need to know where the car started. Two cars with identical speed profiles but different starting points have position functions that differ by a constant. This simple observation is the entire content of the theory of primitives.

DEFINITION **Primitive of a function**

Let f be a function defined on an interval I. A **primitive** of f on I is any differentiable function F such that

$$F'(x) = f(x) \qquad \text{for every } x \in I.$$

We also say that F is an antiderivative of f.

EXEMPLE **Spotting primitives**

1. $F(x) = x^3$ is a primitive of $f(x) = 3x^2$ on $\mathbb{R}$, since $F'(x) = 3x^2$.
2. $F(x) = \sin x$ is a primitive of $f(x) = \cos x$ on $\mathbb{R}$.
3. $F(x) = \ln x$ is a primitive of $f(x) = \dfrac{1}{x}$ on $]0, +\infty[$.

Notice immediately that a primitive is never unique: since the derivative of a constant is zero, if F is a primitive of f, then so is $F + C$ for any constant C. What is remarkable is that there are no other primitives.

PROPERTY **Two primitives differ by a constant**

Let f be a function admitting primitives on an interval I. If F and G are two primitives of f on I, then there exists a constant $C \in \mathbb{R}$ such that

$$G(x) = F(x) + C \qquad \text{for all } x \in I.$$

Consequently, if F is one primitive of f, then all primitives of f are exactly the functions $F + C$, $C \in \mathbb{R}$.

Proof (examinable). Let $H = G - F$. Then H is differentiable on I and

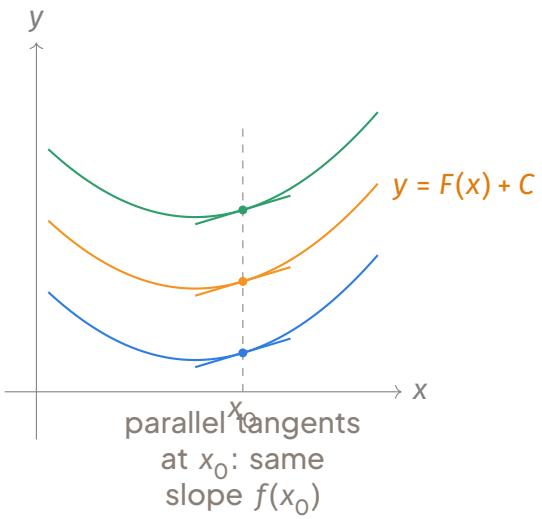
$$H'(x) = G'(x) - F'(x) = f(x) - f(x) = 0 \quad \text{for all } x \in I.$$

A function whose derivative is zero on an interval is constant there (this is a consequence of the Mean Value Theorem: for $a < b$ in I, there exists $c \in]a, b[$ with $H(b) - H(a) = H'(c)(b - a) = 0$, so $H(b) = H(a)$). Hence $H(x) = C$ for some constant C, that is, $G(x) = F(x) + C$. ■

WARNING **The word "interval" matters**

The conclusion fails on a union of disjoint intervals. On $]-\infty, 0[\cup]0, +\infty[$, the function defined by $H(x) = 1$ for $x > 0$ and $H(x) = -1$ for $x < 0$ has derivative 0 everywhere on its domain, yet it is not constant. At the GCE, primitives are always sought on an interval, so you may safely write "$+ C$".

The geometric meaning is beautiful: all the curves $y = F(x) + C$ are vertical translates of one another, and at any fixed abscissa x_0 they all have parallel tangents, since the slope $F'(x_0) = f(x_0)$ does not depend on C.



A family of primitives $y = F(x) + C$: vertical translates with the same tangent slope at each abscissa.

1.2 The indefinite integral

DEFINITION **Indefinite integral**

The **indefinite integral** of f on an interval I is the set of all its primitives. We write

$$\int f(x)\,dx = F(x) + C,$$

where F is any one primitive of f and C is an arbitrary constant, called the **constant of integration**. The symbol $\int$ is the integral sign, f is the integrand, and dx indicates the variable of integration.

The notation $\int f(x)\,dx$ therefore denotes an entire family of functions, not a single function. The constant C is not a decorative detail: it is precisely what distinguishes one primitive from another, and it is what allows us to fit an initial condition later.

WARNING **Never omit the constant C**

In an indefinite integral, forgetting "$+C$" means claiming that f has only one primitive, which is false. At the GCE Advanced Level, a missing constant of integration systematically costs a mark, even if everything else is correct. Write it every single time.

1.3 Table of standard primitives

Every differentiation formula read backwards gives an integration formula. The following table must be known by heart; it is the toolbox for the whole chapter.

Function $f(x)$	A primitive $F(x)$	Valid on
x^n $(n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$	$\mathbb{R}$ (or $\mathbb{R}^*$ if $n < 0$)
$\dfrac{1}{x}$	$\ln\lvert x\rvert$	$]0, +\infty[$ or $]-\infty, 0[$
e^x	e^x	$\mathbb{R}$
$\cos x$	$\sin x$	$\mathbb{R}$
$\sin x$	$-\cos x$	$\mathbb{R}$
$\sec^2 x = \dfrac{1}{\cos^2 x}$	$\tan x$	intervals where $\cos x \neq 0$
$\dfrac{1}{1+x^2}$	$\arctan x$	$\mathbb{R}$
$\dfrac{1}{\sqrt{1-x^2}}$	$\arcsin x$	$]-1, 1[$

WARNING $\ln|x|$, **not** $\ln x$

The primitive of $\frac{1}{x}$ is $\ln|x| + C$. Indeed, for $x < 0$, the derivative of $\ln(-x)$ is $\frac{-1}{-x} = \frac{1}{x}$. Writing $\ln x$ alone excludes all negative values of x and is marked wrong whenever the interval contains negative numbers.

PROPERTY **Linearity of integration**

For any functions f, g admitting primitives on I and any constants $\alpha, \beta \in \mathbb{R}$,

$$\int \big(\alpha f(x) + \beta g(x)\big)\,dx = \alpha \int f(x)\,dx + \beta \int g(x)\,dx.$$

This follows immediately from the linearity of differentiation. In practice: integrate term by term, and pull constants out.

WORKED EXAMPLE 1 **A first indefinite integral**

Find $\int \left(6x^2 - \frac{4}{x} + 3\cos x\right)dx$ on $]0, +\infty[$.

SOLUTION

Solution. By linearity we integrate term by term:

$$\begin{aligned}\int \left(6x^2 - \frac{4}{x} + 3\cos x\right)dx &= 6\int x^2\,dx - 4\int \frac{dx}{x} + 3\int \cos x\,dx \\ &= 6\cdot\frac{x^3}{3} - 4\ln x + 3\sin x + C \\ &= 2x^3 - 4\ln x + 3\sin x + C.\end{aligned}$$

Check: $\frac{d}{dx}\left(2x^3 - 4\ln x + 3\sin x + C\right) = 6x^2 - \frac{4}{x} + 3\cos x$. ✓

1.4 Integration as the reverse of the chain rule

Many integrands are secretly the result of a chain-rule differentiation. Since $\frac{d}{dx}\,G(f(x)) = f'(x)\,G'(f(x))$, any integrand of the form "derivative of the inside × function of the inside" can be integrated at once. Three patterns appear constantly at the GCE.

PROPERTY **Reverse chain rule: three key patterns**

Let f be differentiable on I.

1. Power form: $\int f'(x)\,[f(x)]^n\,dx = \dfrac{[f(x)]^{n+1}}{n+1} + C \quad (n \neq -1)$.
2. Logarithmic form: $\int \dfrac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$ (where f does not vanish).
3. Exponential form: $\int f'(x)\,e^{f(x)}\,dx = e^{f(x)} + C$.

Each formula is proved by differentiating the right-hand side with the chain rule and recovering the integrand.

METHOD **Recognising a reverse chain rule**

1. Identify a "block" $f(x)$ inside the integrand (inside a power, a denominator, an exponential, a square root…).
2. Compute $f'(x)$ and check whether the rest of the integrand is $f'(x)$, possibly up to a constant factor.
3. Adjust the constant: if the integrand contains $k\,f'(x)$ instead of $f'(x)$, divide your answer by k (or factor k out).
4. Apply the matching pattern and add C.
5. Always verify by differentiating your answer.

WORKED EXAMPLE 2 **Reverse chain rule in action**

Determine the following indefinite integrals.

1. $\int 6x\,(x^2+1)^2\,dx$
2. $\int \dfrac{2x+1}{x^2+x+3}\,dx$
3. $\int x\,e^{x^2}\,dx$

SOLUTION

Solution.

1. The block is $f(x) = x^2 + 1$ with $f'(x) = 2x$. The integrand is $3 \cdot 2x\,(x^2+1)^2 = 3f'(x)[f(x)]^2$, so
$$\int 6x\,(x^2+1)^2\,dx = 3 \cdot \frac{(x^2+1)^3}{3} + C = (x^2+1)^3 + C.$$
Check: $\frac{d}{dx}(x^2+1)^3 = 3(x^2+1)^2 \cdot 2x = 6x(x^2+1)^2$. ✓

2. The denominator is $f(x) = x^2+x+3$ and its derivative $f'(x) = 2x+1$ is exactly the numerator. Hence
$$\int \frac{2x+1}{x^2+x+3}\,dx = \ln\left(x^2+x+3\right) + C,$$
where the absolute value may be dropped because $x^2+x+3 > 0$ for all x (its discriminant is $1 - 12 = -11 < 0$).

3. The block is $f(x) = x^2$ with $f'(x) = 2x$; the integrand contains only x, i.e. $\frac{1}{2}f'(x)$. Therefore
$$\int x\,e^{x^2}\,dx = \frac{1}{2}\,e^{x^2} + C.$$
Check: $\frac{d}{dx}\left(\frac{1}{2}e^{x^2}\right) = \frac{1}{2} \cdot 2x\,e^{x^2} = x\,e^{x^2}$. ✓

WARNING **You may only adjust constant factors**

Writing $\int x\,e^{x^2}\,dx = \frac{1}{2x} \cdot 2x\,e^{x^2} \to \frac{e^{x^2}}{2x}$ is a classic blunder: a factor involving x cannot be pulled out of the integral. If the missing factor is not a constant, the reverse chain rule does not apply directly — another technique (substitution, parts) is needed.

1.5 Determining the constant: primitives through a given point

Because the primitives of f form a one-parameter family $F + C$, a single extra condition fixes the primitive uniquely. In applications this condition is physical (initial position, initial charge) or geometric (the curve passes through a given point).

METHOD **Finding the primitive satisfying $F(x_0) = y_0$**

1. Compute the general primitive: $F(x) = G(x) + C$, where G is any particular primitive.
2. Substitute the condition: $G(x_0) + C = y_0$, and solve for C.
3. Write the final answer $F(x) = G(x) + C$ with the value of C found, and check by differentiation plus the condition.

WORKED EXAMPLE 3 **Curve through a given point**

A curve $y = F(x)$ has gradient $\dfrac{dy}{dx} = 3x^2 - \dfrac{2}{x^2}$ for $x > 0$, and passes through the point $(1, 5)$. Find the equation of the curve.

SOLUTION

Solution. Integrate term by term:

$$F(x) = \int \left(3x^2 - \frac{2}{x^2}\right) dx = x^3 + \frac{2}{x} + C.$$

(Recall $\dfrac{-2}{x^2} = -2x^{-2}$ has primitive $-2 \cdot \dfrac{x^{-1}}{-1} = \dfrac{2}{x}$.)
The curve passes through $(1, 5)$, so $F(1) = 5$:

$$1^3 + \frac{2}{1} + C = 5 \implies 3 + C = 5 \implies C = 2.$$

Hence the curve is

$$F(x) = x^3 + \frac{2}{x} + 2.$$

Check: $F'(x) = 3x^2 - \dfrac{2}{x^2}$ ✓ and $F(1) = 1 + 2 + 2 = 5$ ✓.

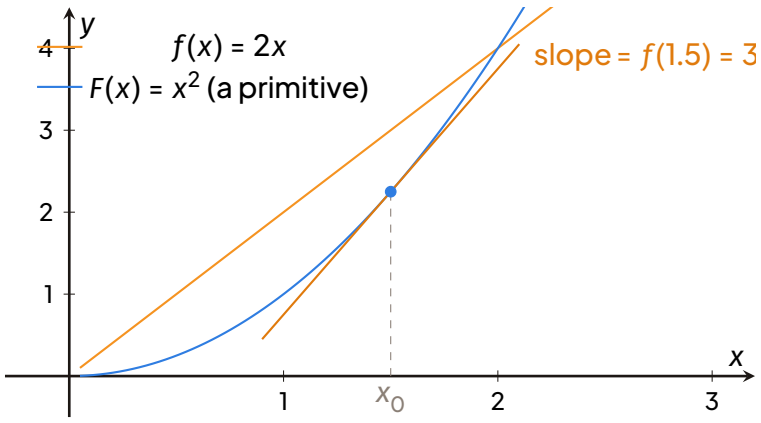
1.6 Verification: differentiate your answer

Integration has a priceless feature that most algebraic manipulations lack: every answer can be checked exactly. Since F is a primitive of f precisely when $F' = f$, differentiating your final answer must return the integrand.

METHOD **The verification reflex**

1. Differentiate your proposed primitive $F(x) + C$ completely (chain rule included).
2. Simplify and compare with the integrand $f(x)$.
3. If they coincide, the integration is correct; if not, locate the discrepancy (usually a missing constant factor) and correct it.

This takes less than a minute in the examination and eliminates the most common source of lost marks.



The function $f(x) = 2x$ and its primitive $F(x) = x^2$: the slope of the tangent to F at x_0 equals the value $f(x_0)$.

KEY POINTS **Key points of this section**

- A **primitive** of f on an interval I is a function F with $F'(x) = f(x)$ for all $x \in I$.
- Two primitives of the same function on an interval differ by a **constant**; hence $\int f(x)\,dx = F(x) + C$.
- The table of standard primitives (powers, $\frac{1}{x}$, e^x, sin, cos, $\sec^2$, $\frac{1}{1+x^2}$, $\frac{1}{\sqrt{1-x^2}}$) must be memorised.
- Integration is **linear**: integrate term by term and pull out constants.
- Master the reverse chain rule: $\int f'[f]^n = \frac{[f]^{n+1}}{n+1}$, $\int \frac{f'}{f} = \ln|f|$, $\int f'e^f = e^f$ (all $+ C$).
- An initial condition $F(x_0) = y_0$ fixes C uniquely.
- Always **verify** an integration by differentiating the answer.

EXERCISE 1 **Practice**

1. Find $\int \left(4x^3 - \frac{3}{x} + \frac{2}{1+x^2}\right) dx$ on $]0, +\infty[$.
2. Find $\int \frac{3x^2}{x^3+5}\,dx$ and $\int \cos x\, e^{\sin x}\,dx$.
3. A curve has gradient $\frac{dy}{dx} = 2x + \frac{1}{\sqrt{x}}$ for $x > 0$ and passes through $(1, 4)$. Find its equation.

SOLUTION **Exercise 1**

1. $\int \left(4x^3 - \frac{3}{x} + \frac{2}{1+x^2}\right) dx = x^4 - 3\ln x + 2\arctan x + C$.
2. $\int \frac{3x^2}{x^3+5}\,dx = \ln|x^3+5| + C$ (numerator = derivative of denominator); $\int \cos x\, e^{\sin x}\,dx = e^{\sin x} + C$ (pattern $\int f' e^f$ with $f = \sin x$).
3. $y = x^2 + 2\sqrt{x} + C$; the condition $y(1) = 4$ gives $1 + 2 + C = 4$, so $C = 1$ and $y = x^2 + 2\sqrt{x} + 1$.

DID YOU KNOW? **From Leibniz to Limbe**

The elongated "S" of the integral sign $\int$ was introduced by Gottfried Wilhelm Leibniz in 1675: it stands for summa, the Latin word for sum, because an integral is the limit of a sum of infinitely many tiny contributions. The same idea powers modern engineering: the flow meters at SONARA essentially compute $\int (\text{rate})\,dt$ continuously to totalise the oil delivered. Every time you write $\int f(x)\,dx = F(x) + C$, you are using a tool that is three and a half centuries old — and still indispensable.

2 DEFINITE INTEGRALS AND THE FUNDAMENTAL THEOREM

In the previous section we learnt how to find primitives: given a function f, we looked for every function F such that $F'(x) = f(x)$, and we wrote the answer as the indefinite integral $\int f(x)\,dx = F(x) + C$. That was a purely algebraic game — reversing differentiation. We now give integration a geometric life: the **definite integral** $\int_a^b f(x)\,dx$ is a number, and that number measures the signed area trapped between the curve $y = f(x)$ and the x-axis. The miracle — one of the most beautiful results in all of mathematics — is that this geometric number is computed using the primitives we have just studied. This is the **Fundamental Theorem of Calculus**.

2.1 From Area to the Definite Integral

An intuitive observation.

Suppose a car travels along the Bamenda–Bafoussam road with constant velocity $v = 60\ \text{km h}^{-1}$ for 2 h. The distance covered is $60 \times 2 = 120$ km — the area of a rectangle under the velocity graph. Now suppose the velocity varies, $v = v(t)$. The distance is no longer a simple product, but we can approximate it: split the time interval into tiny slices of duration Δt; on each slice the velocity is nearly constant, so the distance covered is about $v(t_i)\,\Delta t$, the area of a thin rectangle. Adding all the thin rectangles gives an approximation of the total distance — and of the total area under the curve. As $\Delta t \to 0$, the approximation becomes exact.

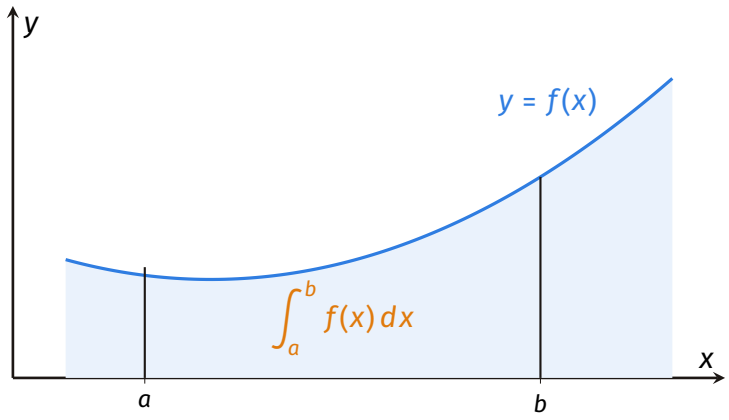
DEFINITION **Definite integral as a limit of a sum**

Let f be continuous on $[a, b]$. Divide $[a, b]$ into n equal subintervals of width $\Delta x = \frac{b-a}{n}$, and choose a point x_i in each subinterval. The **definite integral** of f from a to b is

$$\int_a^b f(x)\,dx = \lim_{n\to+\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x.$$

The symbol $\int$ is an elongated "S" for sum; a and b are the limits of integration; x is a dummy variable: $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$. The limit exists and is independent of the choice of the points x_i because f is continuous.

When $f \geq 0$ on $[a, b]$, this limit is exactly the **area** of the region bounded by the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$.



For $f \geq 0$, the definite integral $\int_a^b f(x)\,dx$ is the area of the shaded region under the curve.

2.2 The Fundamental Theorem of Calculus

Computing limits of sums directly is painful. The Fundamental Theorem replaces all that work by a simple subtraction.

PROPERTY **Fundamental Theorem of Calculus**

Let f be continuous on $[a, b]$ and let F be any primitive of f on $[a, b]$. Then

$$\int_a^b f(x)\,dx = F(b) - F(a), \qquad \text{written } \Big[F(x)\Big]_a^b.$$

Why it is true (instructive sketch).

Define the "area-so-far" function $A(x) = \int_a^x f(t)\,dt$. When x increases by a small amount h, the extra area $A(x+h) - A(x)$ is a thin strip of width h and height about $f(x)$, so $A(x+h) - A(x) \approx f(x)\,h$, i.e.

$$\frac{A(x+h) - A(x)}{h} \approx f(x) \xrightarrow[h \to 0]{} A'(x) = f(x).$$

Thus A is a primitive of f. Since two primitives differ by a constant, $A(x) = F(x) + C$. Now $A(a) = 0$ gives $C = -F(a)$, and therefore

$$\int_a^b f(x)\,dx = A(b) = F(b) - F(a).$$

Note that the constant of integration cancels: $(F(b) + C) - (F(a) + C) = F(b) - F(a)$. This is why any primitive works and why we never write "$+C$" in a definite integral.

EXEMPLE **First computations**

1. $\int_1^3 x^2\,dx = \left[\frac{x^3}{3}\right]_1^3 = \frac{27}{3} - \frac{1}{3} = \frac{26}{3}$.
2. $\int_0^{\pi} \sin x\,dx = \left[-\cos x\right]_0^{\pi} = (-\cos\pi) - (-\cos 0) = 1 + 1 = 2$.
3. $\int_0^2 e^{3x}\,dx = \left[\frac{e^{3x}}{3}\right]_0^2 = \frac{e^6 - 1}{3}$.

WARNING **No "$+C$" in definite integrals**

A definite integral is a number. Writing $\int_0^1 2x\,dx = x^2 + C$ is a classic error: substitute the bounds and the C disappears. Reserve the constant for indefinite integrals only.

2.3 Properties of the Definite Integral

All the properties below follow directly from $\int_a^b f = F(b) - F(a)$.

PROPERTY **Algebraic properties**

Let f, g be continuous and $\lambda, \mu \in \mathbb{R}$.

1. Inversion of bounds: $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$, and $\int_a^a f(x)\,dx = 0$.

2. Linearity: $\int_a^b \big(\lambda f(x) + \mu g(x)\big)\,dx = \lambda \int_a^b f(x)\,dx + \mu \int_a^b g(x)\,dx$.
3. Positivity: if $f(x) \geq 0$ on $[a, b]$ with $a < b$, then $\int_a^b f(x)\,dx \geq 0$. More generally, $f \leq g$ on $[a, b]$ implies $\int_a^b f \leq \int_a^b g$.

PROPERTY **Chasles relation**

For any real numbers a, b, c (in any order, provided f is continuous on an interval containing them),

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Proof (examinable and one line). If F is a primitive of f, then

$$\int_a^c f + \int_c^b f = \big(F(c) - F(a)\big) + \big(F(b) - F(c)\big) = F(b) - F(a) = \int_a^b f. \qquad \blacksquare$$

The Chasles relation is the tool we use to split an integral at points where the behaviour of f changes — for example at the zeros of f when computing areas.

2.4 Area Between a Curve and the x-Axis

Here is the crucial distinction. The integral $\int_a^b f(x)\,dx$ counts area positively where $f > 0$ and negatively where $f < 0$. An area, however, is always a non-negative number.

METHOD **Area between a curve and the x-axis**

To find the area enclosed by $y = f(x)$, the x-axis, $x = a$ and $x = b$:

1. Solve $f(x) = 0$ on $[a, b]$ to locate where the curve crosses the axis.
2. Study the sign of f on each subinterval (a sign table is ideal).
3. Split the integral with the Chasles relation at each zero.
4. Take the absolute value of each piece that comes out negative, then add.

In short: Area $= \int_a^b \big|f(x)\big|\,dx$, computed piecewise.

WORKED EXAMPLE 4 **Area with a sign change**

Find the area of the region bounded by the curve $y = x^2 - 1$, the x-axis, and the lines $x = 0$ and $x = 2$.

SOLUTION

Step 1 — zeros. $x^2 - 1 = 0 \iff x = \pm 1$. On $[0, 2]$, only $x = 1$ matters.
Step 2 — sign. On $[0, 1]$, $x^2 - 1 \le 0$ (curve below the axis); on $[1, 2]$, $x^2 - 1 \ge 0$.
Step 3 — split (Chasles) and take absolute values. A primitive is $F(x) = \dfrac{x^3}{3} - x$.

$$\int_0^1 (x^2 - 1)\,dx = \left[\frac{x^3}{3} - x\right]_0^1 = \frac{1}{3} - 1 = -\frac{2}{3}, \qquad \int_1^2 (x^2 - 1)\,dx = \left(\frac{8}{3} - 2\right) - \left(-\frac{2}{3}\right) = \frac{4}{3}.$$

Step 4 — area.

$$\text{Area} = \left|-\frac{2}{3}\right| + \frac{4}{3} = \frac{6}{3} = 2 \text{ square units.}$$

Remark. The "naive" integral $\displaystyle\int_0^2 (x^2 - 1)\,dx = -\frac{2}{3} + \frac{4}{3} = \frac{2}{3}$ is not the area: the two regions partially cancel. Examiners test exactly this trap.

WARNING **Integral ≠ area**

A definite integral may be negative or zero even when a large region is involved; an area is always ≥ 0. In GCE answers, write clearly: "the integral equals $\frac{2}{3}$" versus "the area equals 2 square units". Confusing the two costs marks every year.

2.5 Area Between Two Curves

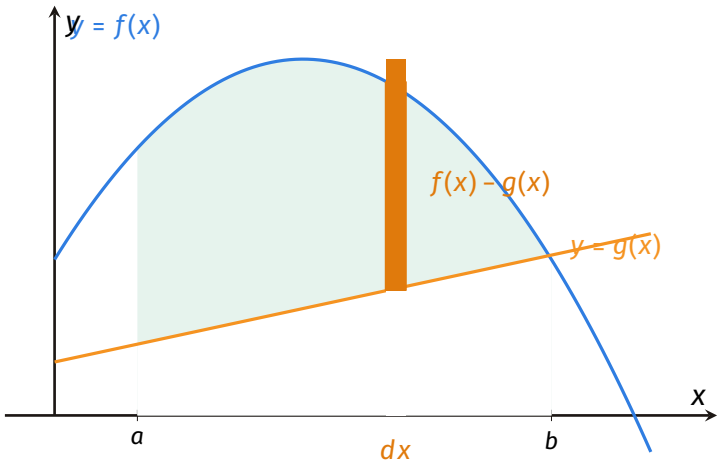
Let f and g be continuous on $[a, b]$ with $f(x) \ge g(x)$. Slice the region between the two curves into thin vertical strips of width dx: each strip is approximately a rectangle of height $f(x) - g(x)$. Summing and passing to the limit:

PROPERTY **Area between two curves**

If $f(x) \ge g(x)$ on $[a, b]$, the area of the region bounded by the two curves and the lines $x = a$, $x = b$ is

$$\text{Area} = \int_a^b \big(f(x) - g(x)\big)\,dx.$$

If the curves cross inside $[a, b]$, split at the intersection points (found by solving $f(x) = g(x)$) so that on each piece the same curve stays on top.



Area between two curves: a strip of width dx has height $f(x) - g(x)$; the integral sums all the strips.

WORKED EXAMPLE 5 **Area between a parabola and a line**

Find the area enclosed by $y = 4 - x^2$ and $y = x + 2$.

SOLUTION

Intersections: $4 - x^2 = x + 2 \iff x^2 + x - 2 = 0 \iff (x - 1)(x + 2) = 0$, so $x = -2$ and $x = 1$.
Which curve is on top? Test $x = 0$: $4 - 0^2 = 4 > 2 = 0 + 2$. So $4 - x^2 \geq x + 2$ on $[-2, 1]$.
Integrate the difference:

$$\text{Area} = \int_{-2}^{1} \left((4 - x^2) - (x + 2)\right) dx = \int_{-2}^{1} \left(2 - x - x^2\right) dx$$

$$= \left[2x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2}^{1} = \left(2 - \frac{1}{2} - \frac{1}{3}\right) - \left(-4 - 2 + \frac{8}{3}\right) = \frac{7}{6} + \frac{10}{3} = \frac{9}{2}.$$

The enclosed area is $\frac{9}{2}$ square units.

2.6 Mean Value of a Function

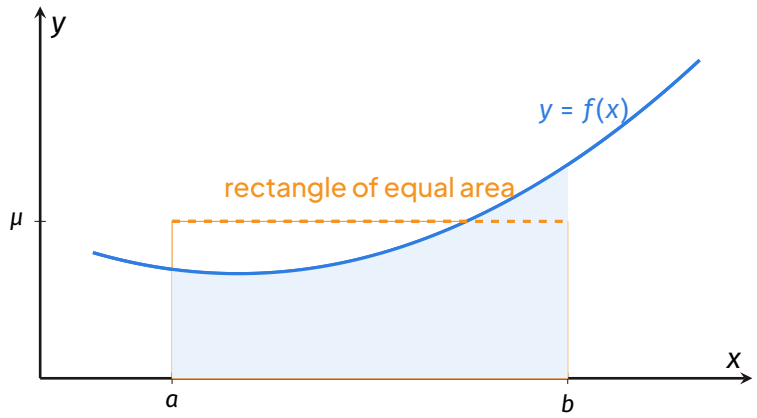
What is the "average temperature" over a day, when the temperature varies continuously? Add up and divide — but continuously.

DEFINITION **Mean value of a function**

The **mean value** of a continuous function f over $[a, b]$ is

$$\mu = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Geometric interpretation. Rewriting, $\mu \times (b-a) = \int_a^b f(x)\,dx$: the rectangle of width $(b-a)$ and height μ has exactly the same area as the region under the curve. The mean value is the "level height" of the function.



The mean value μ is the height of the rectangle over $[a, b]$ whose area equals $\int_a^b f(x)\,dx$.

EXEMPLE **Computing a mean value**

The mean value of $f(x) = x^2$ over $[0, 3]$ is

$$\mu = \frac{1}{3-0}\int_0^3 x^2\,dx = \frac{1}{3}\left[\frac{x^3}{3}\right]_0^3 = \frac{1}{3}\times 9 = 3.$$

Check the intuition: x^2 ranges from 0 to 9 on $[0, 3]$, and 3 is a plausible "level height".

2.7 GCE Extra: Kinematics — Displacement from Velocity

DID YOU KNOW? **Why physicists love the definite integral**

Differentiation turns position into velocity: $v(t) = \frac{ds}{dt}$. Integration reverses this. Since a

primitive of v is the position s, the Fundamental Theorem gives

$$\int_{t_1}^{t_2} v(t)\,dt = s(t_2) - s(t_1) = \text{displacement}.$$

Similarly, $\int_{t_1}^{t_2} a(t)\,dt = v(t_2) - v(t_1)$: the integral of acceleration is the change in velocity. Beware: the total distance travelled is $\int_{t_1}^{t_2} |v(t)|\,dt$ — split at the instants where v changes sign, exactly as for areas.

WORKED EXAMPLE 6 **Displacement and distance**

A particle moves along a straight line with velocity $v(t) = 3t^2 - 12$ (in m s^{-1}, t in s). Find (a) the displacement between $t = 0$ and $t = 3$; (b) the total distance travelled in that time.

SOLUTION

(a) Displacement.

$$s(3) - s(0) = \int_0^3 (3t^2 - 12)\,dt = \left[t^3 - 12t\right]_0^3 = 27 - 36 = -9 \text{ m}.$$

The particle ends 9 m behind its starting point.
(b) Distance. $v(t) = 3(t^2 - 4) = 0$ at $t = 2$ (in $[0, 3]$). On $[0, 2]$, $v < 0$; on $[2, 3]$, $v > 0$.

$$\int_0^2 v\,dt = \left[t^3 - 12t\right]_0^2 = 8 - 24 = -16, \qquad \int_2^3 v\,dt = -9 - (-16) = 7.$$

$$\text{Distance} = |-16| + |7| = 23 \text{ m}.$$

The contrast is striking: displacement −9 m, but distance travelled 23 m — the particle reversed direction at $t = 2$.

EXERCISE 2 **Consolidation**

1. Evaluate: (a) $\int_1^4 \frac{1}{\sqrt{x}}\,dx$; (b) $\int_0^{\pi/2} \cos 2x\,dx$; (c) $\int_{-1}^{2} |x|\,dx$.
2. Find the area bounded by $y = x^2 - 4x + 3$, the x-axis, $x = 0$ and $x = 3$.
3. Find the area enclosed between $y = x^2$ and $y = 2x$.
4. Find the mean value of $f(x) = \sin x$ over $[0, \pi]$.

5. A body has velocity $v(t) = 2t - 6$ m s^{-1}. Find its displacement and the distance travelled between $t = 0$ and $t = 5$ s.

SOLUTION **Exercise 1**

1. (a) $\int_1^4 x^{-1/2}\,dx = \left[2\sqrt{x}\right]_1^4 = 4 - 2 = 2$. (b) $\int_0^{\pi/2} \cos 2x\,dx = \left[\frac{\sin 2x}{2}\right]_0^{\pi/2} = \frac{\sin\pi - \sin 0}{2} = 0$. (c) Split at 0: $\int_{-1}^0 (-x)\,dx + \int_0^2 x\,dx = \frac{1}{2} + 2 = \frac{5}{2}$.

2. $x^2 - 4x + 3 = (x-1)(x-3)$: positive on $[0,1]$, negative on $[1,3]$. With $F(x) = \frac{x^3}{3} - 2x^2 + 3x$: $\int_0^1 = \frac{4}{3}$, $\int_1^3 = -\frac{4}{3}$. Area $= \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$ square units.

3. Intersections: $x^2 = 2x$ gives $x = 0, 2$; line on top. Area $= \int_0^2 (2x - x^2)\,dx = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

4. $\mu = \frac{1}{\pi}\int_0^{\pi} \sin x\,dx = \frac{2}{\pi}$.

5. Displacement: $\left[t^2 - 6t\right]_0^5 = 25 - 30 = -5$ m. Zero of v at $t = 3$: $\int_0^3 = 9 - 18 = -9$, $\int_3^5 = -5 - (-9) = 4$. Distance $= 9 + 4 = 13$ m.

KEY POINTS **Key points of this section**

- $\int_a^b f(x)\,dx$ is a number: the limit of sums of thin rectangles; for $f \geq 0$ it is the area under the curve.
- Fundamental Theorem: $\int_a^b f(x)\,dx = F(b) - F(a)$ for any primitive F — never write $+C$.
- $\int_b^a f = -\int_a^b f$; Chasles: $\int_a^b = \int_a^c + \int_c^b$; linearity; positivity.
- Area $= \int_a^b |f(x)|\,dx$: split at the zeros of f and take absolute values. An integral can be negative; an area cannot.
- Area between two curves: $\int_a^b (f - g)\,dx$ where $f \geq g$; split at intersection points.

- Mean value: $\mu = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$ — the height of the rectangle of equal area.
- Kinematics: displacement $= \displaystyle\int_{t_1}^{t_2} v\,dt$; distance $= \displaystyle\int_{t_1}^{t_2} |v|\,dt$.

3 INTEGRATION OF RATIONAL FUNCTIONS

In the previous section we met the Fundamental Theorem of Calculus, which tells us that evaluating a definite integral amounts to finding any primitive of the integrand. That theorem shifts the whole burden of integration onto one question: how do we actually find primitives? In this section we answer that question completely for one of the most important families of functions in the GCE syllabus: the **rational functions**, that is, quotients of two polynomials,

$$f(x) = \frac{P(x)}{Q(x)}, \qquad P, Q \text{ polynomials.}$$

Rational functions appear everywhere: in the growth of populations, in electric circuits, in the motion of a particle with air resistance. The beautiful news is that every rational function you will meet at Advanced Level can be integrated in closed form, and the answer is always built from only three ingredients: polynomials, logarithms and arctangents. Our job in this section is to learn the machinery that produces them.

3.1 The Two Standard Forms

Everything in this section rests on two standard primitives, which you must know instantly and in both directions.

PROPERTY **Standard primitives for rational functions**

For $a \neq 0$ and x in an interval where the integrand is defined:

$$\int \frac{dx}{x-a} = \ln|x-a| + C, \qquad \int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\!\left(\frac{x}{a}\right) + C.$$

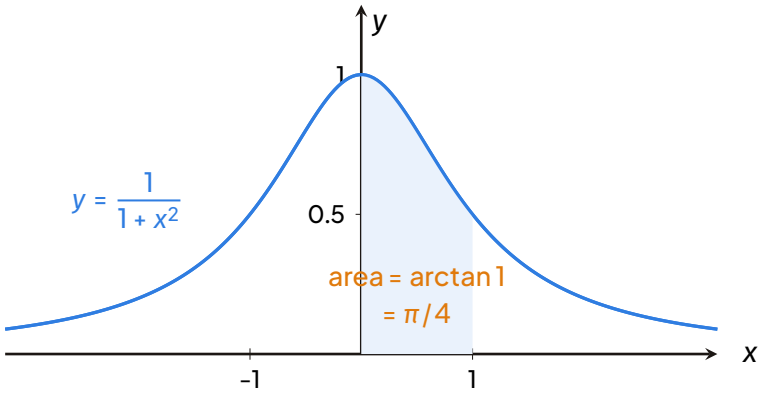
The first formula follows because $\dfrac{d}{dx}\ln|x-a| = \dfrac{1}{x-a}$. For the second, differentiate: with $u = x/a$,

$$\frac{d}{dx}\left[\frac{1}{a}\arctan\!\left(\frac{x}{a}\right)\right] = \frac{1}{a}\cdot\frac{1}{1+(x/a)^2}\cdot\frac{1}{a} = \frac{1}{a^2+x^2}.$$

The absolute value in $\ln|x-a|$ is essential: it makes the formula valid on both sides of the vertical asymptote $x = a$.

WARNING **Do not forget the factor $1/a$**

A very frequent GCE error is to write $\int \frac{dx}{x^2+a^2} = \arctan(x/a) + C$. The chain rule brings out a factor $1/a$, so the correct primitive carries $\frac{1}{a}$ in front. Check by differentiating whenever you are unsure.



The area under $y = \frac{1}{1+x^2}$ from 0 to 1 equals $\arctan 1 = \frac{\pi}{4}$: the arctangent primitive has a direct geometric meaning.

3.2 Step One: Compare the Degrees

Before any clever technique, look at the degrees. If the numerator's degree is greater than or equal to the denominator's, the fraction is improper, and the first move is always **polynomial long division**.

METHOD **Improper fraction? Divide first**

If $\deg P \geq \deg Q$, perform polynomial long division to write

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}, \qquad \deg R < \deg Q.$$

Then integrate the polynomial $S(x)$ term by term, and treat the proper remainder $\frac{R}{Q}$ by the methods below. Never attempt partial fractions on an improper fraction: the decomposition simply does not exist in that form.

EXEMPLE **Long division before integrating**

Compute $I = \int \dfrac{x^2 + 3x + 1}{x + 1}\,dx$.
The numerator has degree 2, the denominator degree 1: we divide. $x^2+3x+1 = (x+1)(x+2)-1$, so

$$\frac{x^2 + 3x + 1}{x + 1} = x + 2 - \frac{1}{x + 1}.$$

Hence

$$I = \int \left(x + 2 - \frac{1}{x + 1}\right) dx = \frac{x^2}{2} + 2x - \ln|x + 1| + C.$$

3.3 Partial Fractions: Distinct Linear Factors

Suppose now the fraction is proper and the denominator factorises into distinct linear factors. Then the fraction splits into a sum of simple pieces, each of which integrates into a logarithm.

PROPERTY **Decomposition into partial fractions**

If $Q(x)$ factorises over $\mathbb{R}$ and $\deg P < \deg Q$, then $\dfrac{P(x)}{Q(x)}$ is a sum of partial fractions whose denominators are the factors of $Q(x)$: a term $\dfrac{A}{x - a}$ for each linear factor, terms $\dfrac{A}{x - a} + \dfrac{B}{(x - a)^2} + \cdots$ for a repeated factor, and a term $\dfrac{Cx + D}{x^2 + px + q}$ for each irreducible quadratic factor. Every one of these pieces has a primitive that is a logarithm, a power, or an arctangent. (The existence of the decomposition is assumed; finding the constants is the examinable skill.)

WORKED EXAMPLE 7 **Distinct linear factors**

Evaluate $I = \int \dfrac{5x - 4}{(x - 1)(x - 2)}\,dx$.

SOLUTION

Step 1: decompose. The fraction is proper with distinct linear factors, so we seek

$$\frac{5x - 4}{(x - 1)(x - 2)} = \frac{A}{x - 1} + \frac{B}{x - 2}.$$

Multiplying through by $(x - 1)(x - 2)$: $5x - 4 = A(x - 2) + B(x - 1)$.
Step 2: find the constants by the cover-up method. Setting $x = 1$: $1 = -A$, so $A = -1$. Setting $x = 2$: $6 = B$.
Step 3: integrate term by term.

$$I = \int \left(\frac{-1}{x - 1} + \frac{6}{x - 2}\right) dx = -\ln|x - 1| + 6\ln|x - 2| + C = \ln\left|\frac{(x - 2)^6}{x - 1}\right| + C.$$

3.4 Repeated Linear Factors and Irreducible Quadratics

WARNING **A repeated factor needs a ladder of terms**

If the denominator contains $(x - a)^2$, the decomposition must contain both $\frac{A}{x-a}$ and $\frac{B}{(x-a)^2}$. Omitting the second term is one of the most common errors at the GCE: with only one constant you will generally find that the identity cannot be satisfied.

WARNING $\frac{1}{(x-a)^2}$ **is a power, not a logarithm**

$$\int \frac{dx}{(x-a)^2} = \int (x-a)^{-2}\,dx = -\frac{1}{x-a} + C.$$

Only the first power $\frac{1}{x-a}$ produces a logarithm. Any higher power integrates as an ordinary power of x.

EXEMPLE **Repeated linear factor**

Compute $I = \int \frac{2x+3}{(x-1)^2}\,dx$.

We write $\frac{2x+3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$, giving $2x + 3 = A(x-1) + B$. Setting $x = 1$: $B = 5$. Comparing coefficients of x: $A = 2$. Therefore

$$I = \int \left(\frac{2}{x-1} + \frac{5}{(x-1)^2}\right) dx = 2\ln|x-1| - \frac{5}{x-1} + C.$$

Notice how the two pieces behave completely differently: one logarithm, one power.

When the denominator contains an irreducible quadratic factor $x^2 + px + q$ (discriminant negative), the corresponding partial fraction has a linear numerator:

$$\frac{Cx + D}{x^2 + px + q}.$$

WORKED EXAMPLE 8 **An irreducible quadratic factor**

Evaluate $I = \int \frac{3x^2 + 2x + 5}{(x+1)(x^2+4)}\,dx$.

SOLUTION

Step 1: decompose. Since $x^2 + 4$ is irreducible, we write

$$\frac{3x^2 + 2x + 5}{(x + 1)(x^2 + 4)} = \frac{A}{x + 1} + \frac{Cx + D}{x^2 + 4}.$$

Multiplying by $(x + 1)(x^2 + 4)$:

$$3x^2 + 2x + 5 = A(x^2 + 4) + (Cx + D)(x + 1).$$

Step 2: constants. Setting $x = -1$: $3 - 2 + 5 = 5A$, so $A = \frac{6}{5}$. Comparing x^2 terms: $3 = A + C$, so $C = \frac{9}{5}$. Comparing constants: $5 = 4A + D$, so $D = \frac{1}{5}$.
Step 3: integrate.

$$I = \frac{6}{5}\ln|x + 1| + \frac{1}{5}\int \frac{9x + 1}{x^2 + 4}\,dx.$$

For the remaining integral, split the numerator into a derivative part and a constant part:

$$\int \frac{9x + 1}{x^2 + 4}\,dx = \frac{9}{2}\int \frac{2x}{x^2 + 4}\,dx + \int \frac{dx}{x^2 + 4} = \frac{9}{2}\ln(x^2 + 4) + \frac{1}{2}\arctan\frac{x}{2}.$$

Finally,

$$I = \frac{6}{5}\ln|x + 1| + \frac{9}{10}\ln(x^2 + 4) + \frac{1}{10}\arctan\frac{x}{2} + C.$$

3.5 The Derivative-of-Denominator Trick

A large proportion of GCE integrals with a quadratic denominator are won or lost on the following observation.

PROPERTY **Numerator equal to the derivative of the denominator**

$$\int \frac{2ax + b}{ax^2 + bx + c}\,dx = \ln\left|ax^2 + bx + c\right| + C,$$

because the integrand has exactly the form $\frac{u'(x)}{u(x)}$ with $u(x) = ax^2 + bx + c$.

METHOD **Split the numerator: derivative part + constant part**

To integrate $\frac{\alpha x + \beta}{ax^2 + bx + c}$:

1. Compute the derivative of the denominator: $(ax^2 + bx + c)' = 2ax + b$.

2. Write $\alpha x + \beta = A(2ax + b) + B$ and identify $A = \dfrac{\alpha}{2a}$, then $B = \beta - Ab$.
3. Split the integral:

$$\int \frac{\alpha x + \beta}{ax^2 + bx + c}\,dx = A \ln\left|ax^2 + bx + c\right| + B \int \frac{dx}{ax^2 + bx + c}.$$

4. Treat the remaining constant-numerator integral by completing the square (below).

3.6 Completing the Square

A constant numerator over a quadratic, $\int \dfrac{dx}{ax^2 + bx + c}$, is handled by **completing the square**. Two cases arise.

Case 1: negative discriminant (no real roots). The quadratic writes as $a\left[(x - h)^2 + k^2\right]$, and the integral becomes an arctangent:

$$\int \frac{dx}{(x - h)^2 + k^2} = \frac{1}{k} \arctan\left(\frac{x - h}{k}\right) + C.$$

Case 2: positive discriminant. The quadratic factorises, and partial fractions give logarithms. Equivalently, completing the square produces a difference of squares, $\int \frac{dx}{(x-h)^2-k^2} = \frac{1}{2k} \ln\left|\frac{x-h-k}{x-h+k}\right| + C$.

EXEMPLE **Completing the square to an arctangent**

Compute $I = \int \dfrac{dx}{x^2 + 4x + 13}$.
Complete the square: $x^2 + 4x + 13 = (x + 2)^2 + 9 = (x + 2)^2 + 3^2$. Hence

$$I = \int \frac{dx}{(x + 2)^2 + 3^2} = \frac{1}{3} \arctan\left(\frac{x + 2}{3}\right) + C.$$

WORKED EXAMPLE 9 **Full strategy: split, then complete the square**

Evaluate $I = \displaystyle\int_0^1 \frac{x + 5}{x^2 + 2x + 5}\,dx$, giving the answer in exact form.

SOLUTION

Step 1: split the numerator. The derivative of the denominator is $2x + 2$, and

$$x + 5 = \tfrac{1}{2}(2x + 2) + 4.$$

Step 2: split the integral.

$$I = \frac{1}{2}\int_0^1 \frac{2x+2}{x^2+2x+5}\,dx + 4\int_0^1 \frac{dx}{x^2+2x+5}.$$

Step 3: logarithmic part.

$$\frac{1}{2}\Big[\ln\left|x^2+2x+5\right|\Big]_0^1 = \frac{1}{2}(\ln 8 - \ln 5) = \frac{1}{2}\ln\frac{8}{5}.$$

Step 4: arctangent part. Since $x^2+2x+5 = (x+1)^2+2^2$,

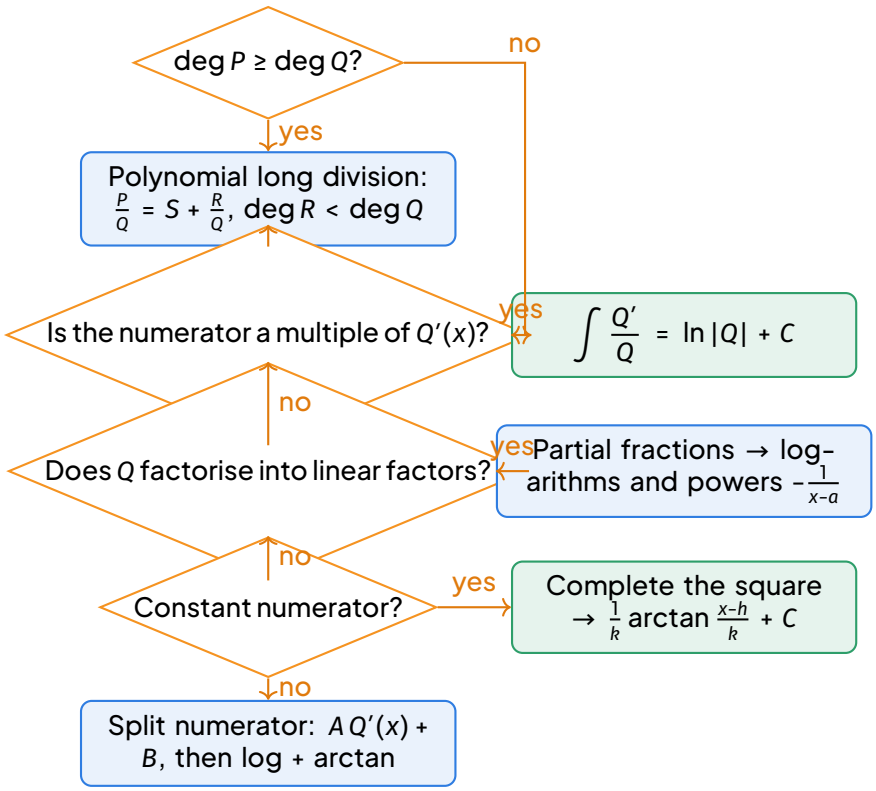
$$4\int_0^1 \frac{dx}{(x+1)^2+4} = 4\cdot\frac{1}{2}\Big[\arctan\frac{x+1}{2}\Big]_0^1 = 2\left(\arctan 1 - \arctan\tfrac{1}{2}\right).$$

Conclusion.

$$I = \frac{1}{2}\ln\frac{8}{5} + 2\left(\frac{\pi}{4} - \arctan\frac{1}{2}\right) = \frac{1}{2}\ln\frac{8}{5} + \frac{\pi}{2} - 2\arctan\frac{1}{2}.$$

3.7 Choosing the Right Treatment: A Strategy

In the examination, no one tells you which method to use. Train yourself to run through the following decision tree every time you meet a rational integrand.



Decision tree for integrating a rational function $\frac{P(x)}{Q(x)}$: compare degrees, divide if necessary, then look for the derivative pattern, factorise, and finish with logarithms, powers or arctangents.

KEY POINTS Key points

- $\int \frac{dx}{x-a} = \ln|x-a| + C$ and $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} + C$.
- If $\deg P \geq \deg Q$, divide first; partial fractions apply only to proper fractions.
- Distinct linear factors give pure logarithms; a repeated factor $(x-a)^2$ needs both $\frac{A}{x-a}$ and $\frac{B}{(x-a)^2}$, and the second integrates to $-\frac{B}{x-a}$, not a logarithm.
- An irreducible quadratic factor takes a linear numerator $\frac{Cx+D}{x^2+px+q}$.
- Numerator = A × (derivative of denominator) + B: the A-part gives a logarithm, the B-part gives an arctangent after completing the square.

WARNING Frequent examination traps

- Forgetting the absolute value in $\ln|x-a|$, or writing $\ln(x^2+4)$ with an absolute value that is unnecessary (a sum of squares is always positive).
- Attempting partial fractions on an improper fraction.
- Writing $\int \frac{dx}{(x-2)^2} = \ln|x-2|^2 + C$: this is wrong; it is a power integral.
- Dropping the factor $\frac{1}{a}$ in the arctangent formula.

EXERCISE 3 **Practice**

1. Compute $\displaystyle\int \frac{3x-1}{(x+1)(x-3)}\,dx$.
2. Compute $\displaystyle\int \frac{x^3}{x-2}\,dx$ (divide first).
3. Compute $\displaystyle\int \frac{4x+1}{x^2+2x+10}\,dx$.
4. Evaluate $\displaystyle\int_1^2 \frac{x+2}{x(x+1)^2}\,dx$.

SOLUTION **Exercises 1–4**

1. Write $\frac{3x-1}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$. Then $3x-1 = A(x-3)+B(x+1)$. Setting $x=-1$: $-4=-4A$, so $A=1$. Setting $x=3$: $8=4B$, so $B=2$. Hence the integral is $\ln|x+1| + 2\ln|x-3| + C$.
2. Long division: $x^3 = (x-2)(x^2+2x+4)+8$, so $\frac{x^3}{x-2} = x^2+2x+4+\frac{8}{x-2}$. Integrating: $\frac{x^3}{3} + x^2 + 4x + 8\ln|x-2| + C$.
3. Derivative of the denominator: $2x+2$. Write $4x+1 = 2(2x+2)-3$. Then

$$\int \frac{4x+1}{x^2+2x+10}\,dx = 2\ln(x^2+2x+10) - 3\int \frac{dx}{(x+1)^2+9} = 2\ln(x^2+2x+10) - \arctan\frac{x+1}{3} + C.$$

4. Decompose: $\frac{x+2}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$. Multiplying out: $x+2 = A(x+1)^2 + Bx(x+1) + Cx$. Setting $x=0$: $A=2$. Setting $x=-1$: $1=-C$, so $C=-1$. Comparing x^2: $0=A+B$, so $B=-2$. Then

$$\int_1^2 \left(\frac{2}{x} - \frac{2}{x+1} - \frac{1}{(x+1)^2}\right)dx = \left[2\ln x - 2\ln(x+1) + \frac{1}{x+1}\right]_1^2 = 2\ln\frac{4}{3} - \frac{1}{6}.$$

With rational functions now under control, we turn next to a technique that dramatically widens the class of functions we can integrate: substitution, together with the elegant shortcuts offered by symmetric integrands.

4 INTEGRATION BY SUBSTITUTION AND SYMMETRY

In the previous section we mastered the integration of rational functions by partial fractions. Many integrals, however, do not present themselves as rational functions. A composite function like $\sin(x^2)\cdot 2x$ or $\dfrac{e^x}{1+e^{2x}}$ can be integrated by recognising an inner function and its derivative. This is the essence of **integration by substitution** (or **change of variable**). We will also see how the **symmetry** of a function over an interval $[-a, a]$ can instantly give the value of a definite integral without any computation.

4.1 The Substitution Rule for Indefinite Integrals

Consider the integral $\int \sin(x^2)\cdot 2x\,dx$. The integrand is a product of a composite function $\sin(x^2)$ and the derivative of the inner function x^2, namely $2x$. If we set $u = x^2$, then $du = 2x\,dx$ and the integral becomes $\int \sin u\,du = -\cos u + C = -\cos(x^2) + C$. This is the **substitution rule** in action: it reverses the chain rule for differentiation.

PROPERTY **Substitution Rule (Indefinite Integrals)**

If $u = g(x)$ is a differentiable function whose range is an interval I, and f is continuous on I, then

$$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du \quad \text{where } u = g(x).$$

In practice we write $du = g'(x)\,dx$ and replace every occurrence of x and dx by u and du.

EXEMPLE **Basic Substitution**

Evaluate $\displaystyle\int \frac{e^x}{1+e^{2x}}\,dx$.

$$\text{Let } u = e^x, \quad du = e^x\,dx.$$

$$\int \frac{e^x}{1+e^{2x}}\,dx = \int \frac{1}{1+u^2}\,du = \arctan u + C = \arctan(e^x) + C.$$

WARNING **The Most Common GCE Error**

When making a substitution, you must replace every x and dx in the integral. Leaving an x inside the integral after writing du is a fatal mistake. Always check: after the substitution, the integrand should contain only u and du.

4.2 Substitution in Definite Integrals: Changing the Limits

For a definite integral $\int_a^b f(g(x))g'(x)\,dx$, we can either find an antiderivative in x and then apply

the limits, or (more efficiently) change the limits directly to u-values and never return to x. The latter method avoids the need to back-substitute and is strongly recommended for GCE examinations.

METHOD **Substitution for Definite Integrals — Change Limits**

1. Choose a substitution $u = g(x)$.
2. Compute $du = g'(x)\,dx$.
3. Change the limits: when $x = a$, $u = g(a)$; when $x = b$, $u = g(b)$.
4. Write the new integral $\int_{g(a)}^{g(b)} f(u)\,du$.
5. Evaluate it directly in terms of u.

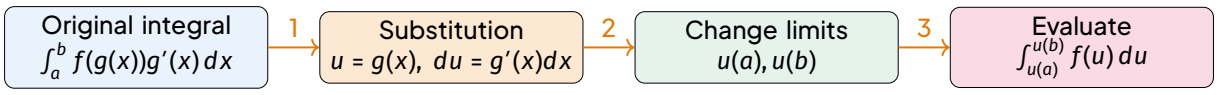
EXEMPLE **Definite Integral with Change of Limits**

Evaluate $\displaystyle\int_0^1 \frac{2x}{1+x^2}\,dx$.

$$\text{Let } u = 1 + x^2, \quad du = 2x\,dx.$$

$$\text{When } x = 0,\ u = 1; \quad \text{when } x = 1,\ u = 2.$$

$$\int_0^1 \frac{2x}{1+x^2}\,dx = \int_1^2 \frac{1}{u}\,du = \left[\ln u\right]_1^2 = \ln 2 - \ln 1 = \ln 2.$$



The substitution pipeline for definite integrals: never substitute back to x.

4.3 Standard Substitutions and Trigonometric Substitutions

Experience teaches us to recognise certain patterns. The table below summarises the most frequent substitutions in GCE examinations.

Integrand contains	Try substitution	Because
$f(ax+b)$	$u = ax+b$	$du = a\,dx$
x^n with $n \neq -1$	$u = x^{n+1}$ or $u = x^n$	simplifies power
e^x or e^{kx}	$u = e^x$	$du = e^x dx$
$\sin x$ or $\cos x$	$u = \sin x$ or $u = \cos x$	derivative is the other
$\tan x$	$u = \tan x$	$du = \sec^2 x\,dx$
$\ln x$	$u = \ln x$	$du = \frac{1}{x}dx$
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$\sqrt{a^2 - a^2\sin^2\theta} = a\cos\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$\sqrt{a^2 + a^2\tan^2\theta} = a\sec\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$\sqrt{a^2\sec^2\theta - a^2} = a\tan\theta$

KEY POINTS **Trigonometric Substitutions — GCE Useful Extra**

The three trigonometric substitutions above are not explicitly listed in the syllabus but appear regularly in GCE papers to integrate expressions involving $\sqrt{a^2 \pm x^2}$. Master them; they turn algebraic square roots into simple trigonometric integrals. For $x = a\sin\theta$, restrict $\theta \in [-\pi/2, \pi/2]$ so that $\cos\theta \geq 0$; for $x = a\tan\theta$, $\theta \in (-\pi/2, \pi/2)$; for $x = a\sec\theta$, $\theta \in [0, \pi/2) \cup (\pi/2, \pi]$ with $\tan\theta \geq 0$ on the relevant branch.

EXEMPLE **Trigonometric Substitution**

Evaluate $\displaystyle\int \frac{dx}{\sqrt{9 - x^2}}$.

Let $x = 3\sin\theta$, $dx = 3\cos\theta\,d\theta$, $\sqrt{9 - x^2} = 3\cos\theta$ $(\cos\theta \geq 0)$.

$$\int \frac{dx}{\sqrt{9 - x^2}} = \int \frac{3\cos\theta}{3\cos\theta}\,d\theta = \int d\theta = \theta + C = \arcsin\left(\frac{x}{3}\right) + C.$$

4.4 Symmetry of Functions: Even and Odd

Before evaluating a definite integral over a symmetric interval $[-a, a]$, always check the **parity** of the integrand.

DEFINITION **Even and Odd Functions**

A function f is even if $f(-x) = f(x)$ for all x in its domain (symmetric about the y-axis).
A function f is odd if $f(-x) = -f(x)$ for all x in its domain (symmetric about the origin).



Left: even function — equal areas on $[-a, 0]$ and $[0, a]$. Right: odd function — signed areas cancel.

PROPERTY **Symmetry Theorem for Definite Integrals**

Let f be integrable on $[-a, a]$.

- If f is even: $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.
- If f is odd: $\int_{-a}^{a} f(x)\,dx = 0$.

EXEMPLE **Using Symmetry**

Evaluate $\int_{-\pi/2}^{\pi/2} \left(x^3 \cos x + \sin x\right) dx$. The function $x^3 \cos x$ is odd (odd × even = odd), $\sin x$ is odd. Sum of odd functions is odd. Hence the integral over $[-\pi/2, \pi/2]$ is 0.

METHOD **Always Test Parity First**

Before computing any definite integral over $[-a, a]$:

1. Check if the integrand is even, odd, or neither.
2. If odd → answer is 0 immediately.
3. If even → compute $2\int_0^a f(x)\,dx$ (often simpler).
4. If neither → proceed with standard methods (substitution, parts, etc.).

4.5 Combining Substitution and Symmetry

Symmetry and substitution are not mutually exclusive; they often work together. A substitution can transform an integral into one over a symmetric interval, or reveal parity that was not obvious in the original variable.

WORKED EXAMPLE 10 **Substitution Reveals Symmetry**

Evaluate $\int_{-1}^{1} \frac{x^3}{\sqrt{1-x^2}}\,dx$. The integrand $f(x) = \frac{x^3}{\sqrt{1-x^2}}$ is odd because $f(-x) = \frac{-x^3}{\sqrt{1-x^2}} = -f(x)$. The interval $[-1, 1]$ is symmetric. Hence

$$\int_{-1}^{1} \frac{x^3}{\sqrt{1-x^2}}\,dx = 0.$$

(No substitution needed; parity alone gives the answer.)

WORKED EXAMPLE 11 **Substitution Creates Symmetric Limits**

Evaluate $\int_{0}^{\pi} \frac{\sin x}{1+\cos^2 x}\,dx$. Let $u = \cos x$, $du = -\sin x\,dx$. When $x = 0$, $u = 1$; when $x = \pi$, $u = -1$.

$$\int_{0}^{\pi} \frac{\sin x}{1+\cos^2 x}\,dx = \int_{1}^{-1} \frac{-du}{1+u^2} = \int_{-1}^{1} \frac{du}{1+u^2}.$$

The new integrand $\frac{1}{1+u^2}$ is even, limits symmetric:

$$\int_{-1}^{1} \frac{du}{1+u^2} = 2\int_{0}^{1} \frac{du}{1+u^2} = 2\big[\arctan u\big]_0^1 = 2\cdot\frac{\pi}{4} = \frac{\pi}{2}.$$

EXEMPLE **Trigonometric Substitution + Symmetry**

Evaluate $\int_{-2}^{2} \sqrt{4-x^2}\,dx$. The integrand $\sqrt{4-x^2}$ is even. So

$$\int_{-2}^{2} \sqrt{4-x^2}\,dx = 2\int_{0}^{2} \sqrt{4-x^2}\,dx.$$

Use $x = 2\sin\theta$, $dx = 2\cos\theta\,d\theta$. When $x = 0$, $\theta = 0$; when $x = 2$, $\theta = \pi/2$.

$$2\int_{0}^{\pi/2} \sqrt{4-4\sin^2\theta}\cdot 2\cos\theta\,d\theta = 2\int_{0}^{\pi/2} 4\cos^2\theta\,d\theta = 8\int_{0}^{\pi/2} \frac{1+\cos 2\theta}{2}\,d\theta = 4\Big[\theta+\frac{\sin 2\theta}{2}\Big]_0^{\pi/2} = 4\cdot\frac{\pi}{2} = 2\pi.$$

(Geometrically this is the area of a semicircle of radius 2: $\frac{1}{2}\pi(2)^2 = 2\pi$.)

4.6 Synthesis and Examination Tips

KEY POINTS **Key Points for GCE**

- Substitution: always change dx and the limits (definite case). Never leave x in the u-integral.
- Standard substitutions: $u = ax + b$, $u = e^x$, $u = \ln x$, $u = \sin x$, $u = \cos x$, $u = \tan x$.
- Trigonometric substitutions: $x = a\sin\theta$ for $\sqrt{a^2 - x^2}$; $x = a\tan\theta$ for $\sqrt{a^2 + x^2}$; $x = a\sec\theta$ for $\sqrt{x^2 - a^2}$.
- Symmetry: test parity first on $[-a, a]$. Odd $\to$ 0; Even $\to$ $2\int_0^a$.
- Combination: a substitution can turn an asymmetric integral into a symmetric one (as in the $\sin x/(1 + \cos^2 x)$ example).

WARNING **Common Traps**

- Forgetting to change the limits in a definite integral and then substituting back to x — this wastes time and often introduces errors.
- Misidentifying parity: $f(x) = x^2 \sin x$ is odd (even × odd), not even.
- Using $x = a\sin\theta$ for $\sqrt{x^2 - a^2}$ (wrong substitution).
- In trigonometric substitution, forgetting to change dx and the limits.

EXERCISE 4 **Practice**

1. Evaluate $\displaystyle\int_0^1 \frac{x}{\sqrt{1 - x^2}}\,dx$.
2. Evaluate $\displaystyle\int_{-\pi}^{\pi} x^2 \sin x\,dx$.
3. Evaluate $\displaystyle\int_0^{\pi/2} \frac{\sin 2x}{1 + \cos^2 x}\,dx$.
4. Evaluate $\displaystyle\int_{-1}^{1} \frac{x^5}{\sqrt{1 - x^2}}\,dx$.
5. Evaluate $\displaystyle\int_0^2 \frac{x^2}{\sqrt{4 - x^2}}\,dx$.

SOLUTION **Exercise 1**

Let $u = 1 - x^2$, $du = -2x\,dx$. When $x = 0$, $u = 1$; when $x = 1$, $u = 0$. $\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx = \int_1^0 \frac{-du/2}{\sqrt{u}} = \frac{1}{2}\int_0^1 u^{-1/2}\,du = \frac{1}{2}[2u^{1/2}]_0^1 = 1$.

SOLUTION **Exercise 2**

$f(x) = x^2 \sin x$ is odd (even × odd). Symmetric interval $[-\pi, \pi] \Rightarrow$ integral = 0.

SOLUTION **Exercise 3**

Let $u = \cos x$, $du = -\sin x\,dx$. $\sin 2x = 2\sin x\cos x = 2\sin x\,u$. When $x = 0$, $u = 1$; when $x = \pi/2$, $u = 0$. $\int_0^{\pi/2} \frac{\sin 2x}{1+\cos^2 x}\,dx = \int_1^0 \frac{2u(-du)}{1+u^2} = 2\int_0^1 \frac{u}{1+u^2}\,du = [\ln(1+u^2)]_0^1 = \ln 2$.

SOLUTION **Exercise 4**

Integrand is odd (odd/even = odd). Symmetric interval $[-1, 1] \Rightarrow 0$.

SOLUTION **Exercise 5**

Even integrand on $[0, 2]$. Use $x = 2\sin\theta$, $dx = 2\cos\theta\,d\theta$. Limits: $0 \to \pi/2$. $\int_0^2 \frac{x^2}{\sqrt{4-x^2}}\,dx = \int_0^{\pi/2} \frac{4\sin^2\theta}{2\cos\theta} \cdot 2\cos\theta\,d\theta = 4\int_0^{\pi/2} \sin^2\theta\,d\theta = 4\int_0^{\pi/2} \frac{1-\cos 2\theta}{2}\,d\theta = 2[\theta - \frac{\sin 2\theta}{2}]_0^{\pi/2} = \pi$.

DID YOU KNOW? **Did You Know?**

The substitution $x = a\tan\theta$ for $\sqrt{a^2 + x^2}$ is the key to deriving the formula for the length of a catenary curve (the shape of a hanging chain). In Cameroon, the cables of the Pont de la Wouri in Douala follow a catenary; engineers use exactly these integrals to compute cable length and tension.

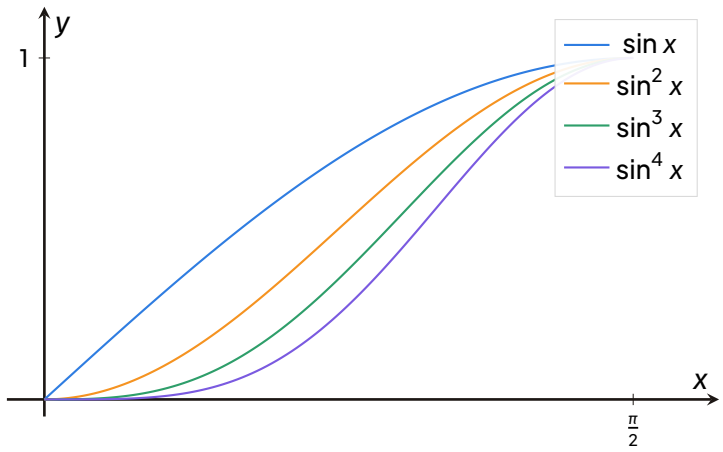
5 REDUCTION FORMULAE AND SYNTHESIS

In the previous section we learnt to simplify integrals by substitution and to exploit symmetry.

But some families of integrals resist every substitution: think of $\int \sin^7 x\,dx$ or $\int x^5 e^x\,dx$. Computing each power separately would be painfully repetitive. The idea of this final section is to be lazy in a clever way: instead of computing each integral from scratch, we find a **recurrence relation** that reduces the power step by step until we reach a trivial integral. This is the technique of **reduction formulae**, a favourite topic of GCE Paper 2, where it is almost always combined with integration by parts. We close the chapter with a synthesis that trains you to choose, at a glance, the right tool among all those studied in this chapter.

5.1 What is a Reduction Formula?

Observation. Suppose you must compute $I_n = \displaystyle\int_0^{\pi/2} \sin^n x\,dx$ for $n = 1, 2, 3, 4$. The graph below shows the curves $y = \sin^n x$ on $[0, \pi/2]$: as n grows, the curve is squeezed towards the axes, so the area under it shrinks regularly. This regular shrinkage suggests that I_n is linked to I_{n-2} by a simple factor — and indeed we shall prove $I_n = \frac{n-1}{n} I_{n-2}$.



The curves $y = \sin^n x$ on $[0, \pi/2]$ for $n = 1, 2, 3, 4$: the area under each curve shrinks regularly as n increases, which is exactly what the reduction formula $W_n = \frac{n-1}{n} W_{n-2}$ captures.

DEFINITION **Reduction formula**

Let $(I_n)_{n\in\mathbb{N}}$ be a family of integrals depending on an integer parameter n, for example $I_n = \int x^n e^x\,dx$. A **reduction formula** is a **recurrence relation** expressing I_n in terms of one or more integrals of lower index, typically I_{n-1} or I_{n-2}:

$$I_n = f(n, x) + g(n)\, I_{n-1} \qquad \text{or} \qquad I_n = f(n, x) + g(n)\, I_{n-2}.$$

Together with an explicitly computed **base case** (I_0 or I_1), the formula allows one to compute I_n for any n by iteration.

The main tool for building reduction formulae is **integration by parts**, which we recall here as

prerequisite consolidation.

KEY POINTS **Integration by parts — reminder**

For differentiable functions u and v,

$$\int u\,\frac{dv}{dx}\,dx = uv - \int v\,\frac{du}{dx}\,dx, \qquad \text{and for definite integrals} \qquad \int_a^b u\,\frac{dv}{dx}\,dx = \Big[uv\Big]_a^b - \int_a^b v\,\frac{du}{dx}\,dx.$$

The golden rule: choose u to be the factor that becomes simpler when differentiated (powers of x, powers of $\ln x$, powers of $\sin x$ or $\cos x$).

METHOD **Building a reduction formula**

1. Write I_n and isolate one power: e.g. $\sin^n x = \sin^{n-1} x \cdot \sin x$.
2. Integrate by parts, choosing u so that differentiation lowers the power (e.g. $u = \sin^{n-1} x$, $dv = \sin x\,dx$).
3. Use identities (e.g. $\cos^2 x = 1 - \sin^2 x$) to make I_n or I_{n-2} reappear on the right-hand side.
4. Solve for I_n: gather the I_n terms on the left and divide by their coefficient.
5. Compute the base case I_0 or I_1 explicitly, then iterate.

5.2 Classical Reduction Formulae

1. The family $I_n = \int x^n e^x\,dx$. Differentiating x^n lowers the power, so take $u = x^n$, $dv = e^x\,dx$, hence $du = nx^{n-1}\,dx$, $v = e^x$:

$$I_n = x^n e^x - n\int x^{n-1}e^x\,dx \quad\Longrightarrow\quad \boxed{I_n = x^n e^x - n\,I_{n-1}}, \qquad I_0 = e^x + C.$$

2. The family $I_n = \int (\ln x)^n\,dx$. Take $u = (\ln x)^n$, $dv = dx$, so $du = \frac{n}{x}(\ln x)^{n-1}\,dx$ and $v = x$:

$$I_n = x(\ln x)^n - n\int (\ln x)^{n-1}\,dx \quad\Longrightarrow\quad \boxed{I_n = x(\ln x)^n - n\,I_{n-1}}, \qquad I_0 = x + C.$$

3. The family $I_n = \int \sin^n x\,dx$ ($n \geq 2$). Write $\sin^n x = \sin^{n-1} x \cdot \sin x$ and take $u = \sin^{n-1} x$, $dv = \sin x\,dx$, so $du = (n-1)\sin^{n-2} x \cos x\,dx$, $v = -\cos x$:

$$\begin{aligned} I_n &= -\cos x\,\sin^{n-1} x + (n-1)\int \sin^{n-2} x \cos^2 x\,dx \\ &= -\cos x\,\sin^{n-1} x + (n-1)\int \sin^{n-2} x\,(1-\sin^2 x)\,dx \\ &= -\cos x\,\sin^{n-1} x + (n-1)I_{n-2} - (n-1)I_n. \end{aligned}$$

Bringing $(n-1)I_n$ to the left: $nI_n = -\cos x \sin^{n-1} x + (n-1)I_{n-2}$, hence

$$\boxed{I_n = -\frac{\cos x \, \sin^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}}.$$

4. The family $I_n = \int \cos^n x \, dx$ ($n \geq 2$). The identical manipulation with $\cos^n x = \cos^{n-1} x \cdot \cos x$ (taking $u = \cos^{n-1} x$, $dv = \cos x \, dx$, so $v = \sin x$) gives

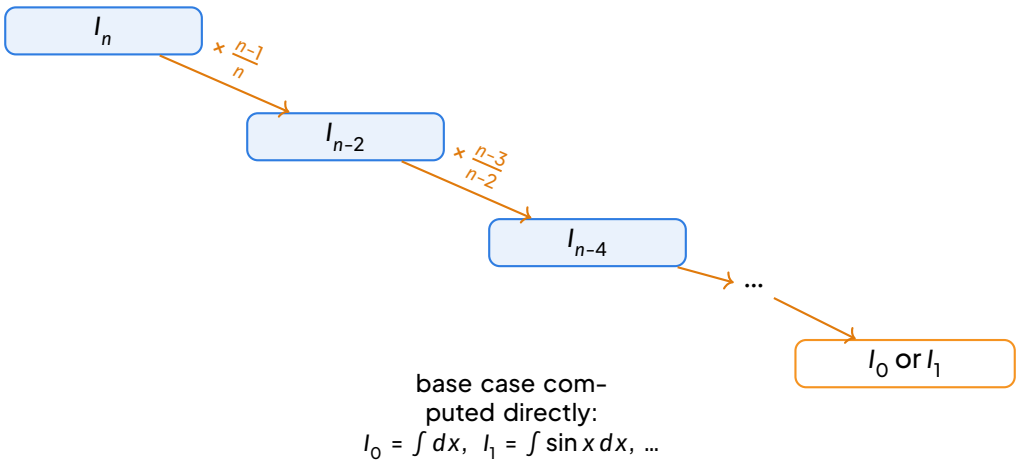
$$\boxed{I_n = \frac{\sin x \, \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}}.$$

WARNING **Show the integration by parts step!**

In the GCE, reduction formulae are usually given in the question or asked to be proved. When asked to prove one, write the integration by parts step explicitly: state your u, dv, du, v, apply the formula, then use the trigonometric identity and solve for I_n. Method marks are awarded for these intermediate lines — jumping straight to the boxed result earns almost nothing.

5.3 Iterating Down to the Base Case

A reduction formula is a ladder: each application moves you down one or two rungs, until you land on I_0 or I_1, which you must compute directly.



The reduction ladder: each step lowers the index (here by 2), until the base case I_0 or I_1 is reached.

METHOD **Using a reduction formula cleanly**

1. State the base case first: compute I_0 (and I_1 if the step is -2) before any iteration. An incorrect base case ruins the whole chain: every subsequent value inherits the error. Examiners award a specific mark for a correct I_0 or I_1 — claim it.
2. Apply the formula step by step, writing one line per reduction; do not skip rungs.
3. Substitute back progressively, keeping exact fractions.
4. For indefinite integrals, do not forget the constant C at the very end.

WORKED EXAMPLE 12 **Computing $\int x^3 e^x \, dx$ by reduction**

Let $I_n = \int x^n e^x \, dx$. Given $I_n = x^n e^x - n I_{n-1}$, compute $\int x^3 e^x \, dx$.

SOLUTION

Solution. Base case: $I_0 = \int e^x \, dx = e^x$. Iterating:

$$I_1 = xe^x - I_0 = xe^x - e^x,$$
$$I_2 = x^2 e^x - 2I_1 = x^2 e^x - 2xe^x + 2e^x,$$
$$I_3 = x^3 e^x - 3I_2 = x^3 e^x - 3x^2 e^x + 6xe^x - 6e^x.$$

Hence $\int x^3 e^x \, dx = e^x(x^3 - 3x^2 + 6x - 6) + C$. Check (good habit): differentiating $e^x(x^3 - 3x^2 + 6x - 6)$ gives $e^x(x^3 - 3x^2 + 6x - 6) + e^x(3x^2 - 6x + 6) = x^3 e^x$. ✓

WORKED EXAMPLE 13 **Computing $\int \sin^4 x \, dx$ by reduction**

Using $I_n = -\dfrac{\cos x \, \sin^{n-1} x}{n} + \dfrac{n-1}{n} I_{n-2}$ for $I_n = \int \sin^n x \, dx$, find $\int \sin^4 x \, dx$.

SOLUTION

Solution. Base case: $I_0 = \int dx = x$. Then

$$I_2 = -\frac{\cos x \, \sin x}{2} + \frac{1}{2} I_0 = -\frac{\cos x \, \sin x}{2} + \frac{x}{2},$$

$$I_4 = -\frac{\cos x \, \sin^3 x}{4} + \frac{3}{4} I_2 = -\frac{\cos x \, \sin^3 x}{4} - \frac{3 \cos x \sin x}{8} + \frac{3x}{8}.$$

Hence $\int \sin^4 x \, dx = -\frac{1}{4} \cos x \sin^3 x - \frac{3}{8} \cos x \sin x + \frac{3}{8} x + C$, which agrees with the result obtained by linearising $\sin^4 x$ with double-angle formulas.

5.4 Definite-Integral Reduction: the Wallis Integrals

When the integral is taken over $\left[0, \frac{\pi}{2}\right]$, the boundary term in the reduction of $\int \sin^n x\,dx$ vanishes (since $\cos\frac{\pi}{2} = 0$ and $\sin 0 = 0$), leaving a beautifully simple relation.

PROPERTY **Wallis integrals**

Let $W_n = \int_0^{\pi/2} \sin^n x\,dx$ for $n \in \mathbb{N}$. Then

$$W_n = \frac{n-1}{n} W_{n-2} \quad (n \geq 2), \qquad W_0 = \frac{\pi}{2}, \qquad W_1 = 1.$$

Consequently:

- if $n = 2p$ is even: $W_{2p} = \dfrac{(2p-1)(2p-3)\cdots 3\cdot 1}{(2p)(2p-2)\cdots 4\cdot 2}\cdot\dfrac{\pi}{2}$;
- if $n = 2p+1$ is odd: $W_{2p+1} = \dfrac{(2p)(2p-2)\cdots 4\cdot 2}{(2p+1)(2p-1)\cdots 5\cdot 3}$.

The same values hold for $\int_0^{\pi/2} \cos^n x\,dx$ (by the substitution $x \mapsto \frac{\pi}{2} - x$, which swaps sin and cos while preserving the interval).

Proof (examinable). From the reduction for $I_n = \int \sin^n x\,dx$ established above, evaluating between 0 and $\frac{\pi}{2}$:

$$\left[-\cos x\,\sin^{n-1} x\right]_0^{\pi/2} = \left(-0\cdot 1\right) - \left(-1\cdot 0\right) = 0, \qquad \text{so} \qquad W_n = \frac{n-1}{n} W_{n-2}.$$

For n even, iterating down to $W_0 = \int_0^{\pi/2} dx = \frac{\pi}{2}$ gives the even formula; for n odd, iterating down to $W_1 = \int_0^{\pi/2} \sin x\,dx = [-\cos x]_0^{\pi/2} = 1$ gives the odd formula. ■

EXEMPLE **Two Wallis computations**

$W_4 = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{3\pi}{16}$ (even, ends with $\frac{\pi}{2}$); $W_5 = \frac{4}{5}\cdot\frac{2}{3}\cdot 1 = \frac{8}{15}$ (odd, ends with 1).

WARNING **Even vs odd — the classic trap**

The even chain ends at $W_0 = \frac{\pi}{2}$, so the answer contains π; the odd chain ends at $W_1 = 1$, so the answer is a plain fraction. Students who stop the chain one step too early (or too late) systematically get the wrong form. Always check: does my answer contain π exactly when n is even?

DID YOU KNOW? **Wallis and π**

The English mathematician John Wallis used these very integrals in 1655 to obtain his famous infinite product $\frac{\pi}{2} = \frac{2\cdot2\cdot4\cdot4\cdot6\cdot6\cdots}{1\cdot3\cdot3\cdot5\cdot5\cdot7\cdots}$, one of the first exact formulas for π — a beautiful example of how a simple recurrence can unlock deep mathematics.

5.5 Synthesis: Choosing the Right Technique in the Exam

GCE Paper 2 integration questions rarely announce their method. Train yourself to run this checklist:

What you see	First technique to try
Polynomial, exponential, trig, ln alone	Direct integration (standard primitives)
Rational function $\frac{P(x)}{Q(x)}$	Degree check, then partial fractions
$f(g(x))\,g'(x)$, or $\sqrt{a^2-x^2}$, $\frac{1}{a^2+x^2}$	Substitution ($u = g(x)$, $x = a\sin\theta$, $x = a\tan\theta$)
Odd/even integrand on $[-a, a]$	Symmetry: odd $\Rightarrow$ 0; even $\Rightarrow$ $2\int_0^a$
Power n in the integrand, family I_n	Reduction formula via integration by parts
Product $x^n e^x$, $x^n \sin x$, $x^n \ln x$	Integration by parts (once or via reduction)

WORKED EXAMPLE 14 **Exam-style synthesis**

(a) Evaluate $\int_{-1}^{1} x^3\cos^2 x\,dx$. (b) Given that $W_n = \int_0^{\pi/2}\cos^n x\,dx$ satisfies $W_n = \frac{n-1}{n}W_{n-2}$, evaluate $\int_0^{\pi/2}\cos^6 x\,dx$.

SOLUTION

Solution. (a) x^3 is odd and $\cos^2 x$ is even, so the product $x^3\cos^2 x$ is odd on the symmetric interval $[-1, 1]$: the integral equals $\boxed{0}$ — no computation needed. (b) $n = 6$ is even, so the chain ends at $W_0 = \frac{\pi}{2}$:

$$W_6 = \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot W_0 = \frac{15}{48}\cdot\frac{\pi}{2} = \boxed{\frac{5\pi}{32}}.$$

EXERCISE 5 **Practice**

1. Prove that $I_n = \int(\ln x)^n\,dx$ satisfies $I_n = x(\ln x)^n - nI_{n-1}$, and deduce $\int(\ln x)^2\,dx$.
2. Using $W_n = \frac{n-1}{n}W_{n-2}$, compute W_3, W_6 and $\int_0^{\pi/2}\sin^7 x\,dx$.

3. Evaluate $\int_{-\pi/4}^{\pi/4} \sin^5 x\,dx$ in one line, justifying your answer.

4. Let $I_n = \int_0^1 x^n e^x\,dx$. Show that $I_n = e - nI_{n-1}$, and deduce the exact value of I_2.

SOLUTION **Exercise 1**

1. $u = (\ln x)^n$, $dv = dx$ give $du = \frac{n}{x}(\ln x)^{n-1}dx$, $v = x$, so $I_n = x(\ln x)^n - n\int(\ln x)^{n-1}dx$. Then $I_2 = x(\ln x)^2 - 2\big(x\ln x - x\big) = x(\ln x)^2 - 2x\ln x + 2x + C$. 2. $W_3 = \frac{2}{3}\cdot 1 = \frac{2}{3}$; $W_6 = \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{5\pi}{32}$; $W_7 = \frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot 1 = \frac{16}{35}$. 3. $\sin^5 x$ is odd and the interval $[-\frac{\pi}{4}, \frac{\pi}{4}]$ is symmetric about 0, so the integral is 0. 4. By parts with $u = x^n$, $dv = e^x dx$ on $[0,1]$: $I_n = \big[x^n e^x\big]_0^1 - nI_{n-1} = e - nI_{n-1}$. Base case $I_0 = e - 1$; then $I_1 = e - (e-1) = 1$ and $I_2 = e - 2$.

WARNING **Time-management and presentation tips for Paper 2**

- If a reduction formula is given, do not re-prove it unless asked — quote it and iterate; save your time.
- Prove it only when the question says "show that": then write the full by-parts step.
- Keep fractions exact until the final line; premature decimals cost accuracy marks.
- Budget roughly 2 minutes per mark: a 6-mark reduction question deserves about 12 minutes, no more.
- When stuck on a definite integral, check symmetry before attempting heavy computation — it may be zero.

KEY POINTS **Key points of the section**

- A reduction formula links I_n to I_{n-1} or I_{n-2}; it is built by integration by parts, choosing u so that differentiation lowers the power.
- Core formulas: $\int x^n e^x\,dx$: $I_n = x^n e^x - nI_{n-1}$; $\int(\ln x)^n\,dx$: $I_n = x(\ln x)^n - nI_{n-1}$; $\int \sin^n x\,dx$: $I_n = -\frac{\cos x \sin^{n-1} x}{n} + \frac{n-1}{n}I_{n-2}$; $\int \cos^n x\,dx$: $I_n = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n}I_{n-2}$.
- Wallis: $W_n = \frac{n-1}{n}W_{n-2}$, with $W_0 = \frac{\pi}{2}$, $W_1 = 1$; even n gives a multiple of π, odd n a plain fraction.
- Always compute and state the base case first, iterate line by line, and check the final

answer by differentiation when possible.

6 INTEGRATION ACTIVITY

6.1 Aims of the activity

This activity is a **modelling task** that brings together, in one coherent problem, almost every skill of the chapter Further Integration. By the end of the group work, you should be able to:

- recognise and **prove** the parity (odd or even character) of a function, and use it to simplify a definite integral over a symmetric interval;
- distinguish carefully between a **signed integral** $\int_a^b f(x)\,dx$ and a **geometric area**, and explain the difference in a concrete situation;
- carry out a **simple substitution** (here $u = x^2 + 4$) to find a primitive of a rational function of the form $\frac{u'}{u}$;
- compute the **mean value** of a function on an interval and interpret it physically;
- use the standard primitive $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a} + C$ in a real context;
- present a complete, rigorous chain of reasoning orally and in writing, as expected at the GCE Advanced Level.

KEY POINTS **Skills mobilised**

Parity of functions → symmetry of integrals → substitution $u = x^2 + 4$ → primitives of rational functions → mean value → arctangent form. One single situation, five lessons of the chapter.

6.2 Situation

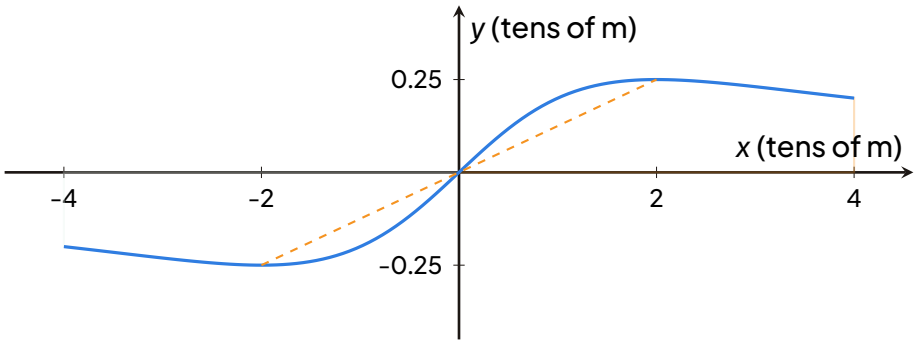
The Nkam River crossing.

The Nkam river, in the Littoral Region, is crossed daily by heavy traffic between Douala and the West Region. As part of the rehabilitation of a bridge over the river, a contractor must build an **approach ramp**: an embankment of compacted laterite that rises gently from the flood plain on one side, crosses the abutment, and descends on the other side.

The design engineer models the profile (the centre-line elevation) of the ramp, in a coordinate system whose origin is at the abutment, by the curve

$$y = f(x) = \frac{x}{x^2 + 4}, \qquad x \in [-4,\, 4],$$

where the unit on both axes is **ten metres** (so $x = 4$ corresponds to 40 m from the abutment, and $y = 1$ to a height of 10 m). Negative values of y correspond to a cutting (the ground must be excavated) and positive values to an embankment (fill must be brought in).



Profile of the approach ramp $y = \frac{x}{x^2+4}$: excavation on $[-4, 0]$ (green), embankment on $[0, 4]$ (orange).

The contractor needs answers to three practical questions.

1. **Earthworks balance.** A quick calculation of $\int_{-4}^{4} f(x)\,dx$ seems to give zero. The site manager concludes: "the excavated soil exactly compensates the fill, so no laterite needs to be purchased." Is he right?
2. **Facing wall.** The side face of the ramp (the vertical wall following the profile, on $[0, 4]$) must be covered with masonry. What is its true area?
3. **Drainage.** A drainage channel running along the ramp has a cross-section modelled by $g(x) = \frac{1}{x^2+4}$ for $x \in [-2, 2]$ (same units). What is the area of this cross-section?

All numerical answers should be given exactly, then rounded to 3 decimal places, and finally converted into **real units** (square metres, metres).

6.3 Tasks

Work in groups of three or four. Write up every step: at the GCE Advanced Level, the reasoning is marked, not only the final number.

Part A — Parity and the signed integral.

1. Show that for all $x \in [-4, 4]$, $f(-x) = -f(x)$. What do we call such a function? What is the symmetry of its graph?
2. State the symmetry property of definite integrals for an odd function on $[-a, a]$, and deduce the value of $I = \int_{-4}^{4} \frac{x}{x^2+4}\,dx$.
3. Explain, in two or three sentences addressed to the site manager, why a zero signed integral does **not** mean that the volumes of excavation and fill cancel out in practice. What quantity should have been computed instead?

Part B — The true area of the side face.

4. Justify that the area of the side face on $[-4, 4]$ is $A = \int_{-4}^{4} \left| \frac{x}{x^2+4} \right| dx$, and use parity to reduce this to $A = 2\int_{0}^{4} \frac{x}{x^2+4}\,dx$.

5. Compute $J = \int_{0}^{4} \frac{x}{x^2+4}\,dx$ using the substitution $u = x^2+4$ (remember to change the bounds). Recognise the form $\frac{u'}{u}$.

6. Deduce A exactly, then in m^2 (beware of the unit: one unit of area on the graph equals 10 m × 10 m).

Part C — Mean height of the ramp.

7. Recall the formula for the mean value of a continuous function on $[a, b]$. Compute the mean height $\bar{y}$ of the ramp on $[0, 4]$, and convert it to metres. Give a one-sentence physical interpretation for the contractor.

Part D — Extension: the drainage channel.

8. Show that g is even and reduce $K = \int_{-2}^{2} \frac{dx}{x^2+4}$ to an integral over $[0, 2]$.

9. Using $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} + C$ with $a = 2$, compute K exactly (a multiple of π is expected), then in m^2.

10. **Synthesis.** Prepare a five-minute board presentation: parity argument, substitution, mean value, arctangent result, and the answer to the site manager's error.

WARNING **Pitfalls to avoid**

- Confusing $\int_{-a}^{a} f$ (signed) with $\int_{-a}^{a} |f|$ (area). An odd function has a zero signed integral but generally a non-zero area.
- Forgetting to change the bounds when substituting $u = x^2 + 4$.
- Writing $\ln(x^2+4)$ without absolute value is acceptable here since $x^2 + 4 > 0$, but say so explicitly.
- Forgetting the factor $\frac{1}{2}$ coming from $du = 2x\,dx$.
- Giving the area in "square units" and forgetting the conversion: 1 square unit = $100\ \text{m}^2$.

6.4 Model answer

Part A.

1. For all $x \in [-4, 4]$,

$$f(-x) = \frac{-x}{(-x)^2 + 4} = \frac{-x}{x^2 + 4} = -f(x).$$

Hence f is an **odd function**; its graph is symmetric with respect to the **origin** of the coordinate system (central symmetry).

2. Symmetry property: if f is odd and continuous on $[-a, a]$, then $\int_{-a}^{a} f(x)\,dx = 0$. (Proof: with $x = -t$ on $[-a, 0]$, $\int_{-a}^{0} f(x)\,dx = \int_{a}^{0} f(-t)(-dt) = -\int_{0}^{a} f(t)\,dt$.) Therefore

$$I = \int_{-4}^{4} \frac{x}{x^2 + 4}\,dx = 0.$$

3. The signed integral counts the region below the axis ($x < 0$, excavation) negatively and the region above ($x > 0$, fill) positively; the two contributions cancel algebraically. Physically, however, excavation and embankment are two separate quantities of material, both positive: excavated soil is rarely reusable as compacted structural fill, and in any case the contractor must pay for both operations. The relevant quantity is the integral of $|f|$, not of f.

Part B.

4. On $[-4, 0]$, $f(x) \le 0$; on $[0, 4]$, $f(x) \ge 0$. The geometric area is therefore $A = \int_{-4}^{4} |f(x)|\,dx$. Since f is odd, $|f|$ is **even** ($|f(-x)| = |-f(x)| = |f(x)|$), so by symmetry of even functions:

$$A = 2\int_{0}^{4} \frac{x}{x^2 + 4}\,dx.$$

5. Let $u = x^2 + 4$, so $du = 2x\,dx$, i.e. $x\,dx = \frac{du}{2}$. Bounds: $x = 0 \Rightarrow u = 4$; $x = 4 \Rightarrow u = 20$. Then

$$J = \int_{0}^{4} \frac{x\,dx}{x^2 + 4} = \frac{1}{2}\int_{4}^{20} \frac{du}{u} = \frac{1}{2}\Big[\ln u\Big]_{4}^{20} = \frac{1}{2}\ln\frac{20}{4} = \frac{1}{2}\ln 5.$$

(Note: $u > 0$ on $[4, 20]$, so the absolute value inside the logarithm is not needed.)

6. Hence

$$A = 2J = \ln 5 \approx 1.609 \text{ square units.}$$

Converting: $A = 100 \ln 5 \approx 160.944\ \text{m}^2$ of masonry facing.

Part C.

7. The mean value of f on $[0, 4]$ is

$$\bar{y} = \frac{1}{4 - 0}\int_{0}^{4} f(x)\,dx = \frac{J}{4} = \frac{\ln 5}{8} \approx 0.201 \text{ units,}$$

i.e. $\bar{y} \approx 2.012$ m. Interpretation: a rectangular embankment of constant height 2.01 m over the 40 m stretch would present the same side area as the real curved ramp.

Part D.

8. $g(-x) = \frac{1}{(-x)^2 + 4} = g(x)$: g is **even**, so

$$K = \int_{-2}^{2} \frac{dx}{x^2 + 4} = 2\int_{0}^{2} \frac{dx}{x^2 + 4}.$$

9. With $a = 2$, a primitive is $\frac{1}{2}\arctan\frac{x}{2}$. Thus

$$K = 2\left[\frac{1}{2}\arctan\frac{x}{2}\right]_0^2 = \arctan 1 - \arctan 0 = \frac{\pi}{4} \approx 0.785 \text{ square units},$$

that is $K = 25\pi \approx 78.540\ \text{m}^2$ for the channel cross-section.

10. Synthesis (expected presentation outline).

- f odd $\Rightarrow$ signed integral zero, but area uses $|f|$ — the site manager's conclusion is wrong;
- substitution $u = x^2 + 4$ turns $\frac{x}{x^2+4}$ into $\frac{1}{2}\cdot\frac{u'}{u}$, giving logarithms;
- mean height $= \frac{1}{b-a}\int_a^b f$;
- $\frac{1}{x^2+a^2}$ integrates to an arctangent, giving the exact value $\frac{\pi}{4}$.

DID YOU KNOW? **Did you know?**

The identity $\arctan 1 = \frac{\pi}{4}$ used above is the simplest case of the Machin-type formulae, once used to compute thousands of digits of π by hand. Every time a GCE candidate integrates $\frac{1}{x^2+a^2}$, a little piece of π appears — a reminder that the arctangent function is the bridge between rational functions and the geometry of the circle.

7 METHODS AND WORKED EXAMPLES

8 PRIMITIVES AND THE INDEFINITE INTEGRAL

DID YOU KNOW? **Historical Note**

The notation $\int f(x)\,dx$ for the indefinite integral was introduced by Gottfried Wilhelm Leibniz in 1675. The elongated 'S' stands for summa (sum), reflecting the integral's origin as a limit of sums. The constant of integration C was emphasised by Euler, who recognised that differentiation loses constant information.

DEFINITION **Primitive and Indefinite Integral**

A function F is a **primitive** (or **antiderivative**) of f on an interval I if $F'(x) = f(x)$ for all $x \in I$. The **indefinite integral** of f is the family of all its primitives, denoted

$$\int f(x)\,dx = F(x) + C, \qquad C \in \mathbb{R}.$$

The function f is the **integrand**, x is the **variable of integration**, and C is the **arbitrary constant**.

PROPERTY **Linearity of the Indefinite Integral**

For any functions f, g integrable on an interval and constants $a, b \in \mathbb{R}$,

$$\int \big[af(x) + bg(x)\big]\,dx = a\int f(x)\,dx + b\int g(x)\,dx.$$

Proof: Differentiate the right-hand side using linearity of the derivative.

KEY POINTS **Table of Standard Primitives (Memorise)**

Function $f(x)$	Primitive $\int f(x)\,dx$
$x^n\ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$\frac{1}{x}$	$\ln\lvert x\rvert + C$
e^{ax}	$\frac{1}{a}e^{ax} + C$
$a^x\ (a > 0, a \neq 1)$	$\frac{a^x}{\ln a} + C$
$\sin(ax)$	$-\frac{1}{a}\cos(ax) + C$
$\cos(ax)$	$\frac{1}{a}\sin(ax) + C$
$\sec^2(ax)$	$\frac{1}{a}\tan(ax) + C$
$\csc^2(ax)$	$-\frac{1}{a}\cot(ax) + C$
$\sec(ax)\tan(ax)$	$\frac{1}{a}\sec(ax) + C$
$\csc(ax)\cot(ax)$	$-\frac{1}{a}\csc(ax) + C$
$\frac{1}{a^2 + x^2}$	$\frac{1}{a}\arctan\frac{x}{a} + C$
$\frac{1}{\sqrt{a^2 - x^2}}$	$\arcsin\frac{x}{a} + C$
$\frac{1}{x\sqrt{x^2 - a^2}}$	$\frac{1}{a}\operatorname{arcsec}\frac{\lvert x\rvert}{a} + C$

METHOD **Finding Primitives: Standard Forms and Linearity**

Goal: Determine $\int f(x)\,dx = F(x) + C$ where $F'(x) = f(x)$.
Steps:

1. Identify the form: Match the integrand to a standard derivative result (power, exponential, trigonometric, logarithmic, inverse trigonometric).
2. Apply Linearity: Split sums/differences; pull out constants.
3. Adjust for Chain Rule (Mental Substitution): If the integrand is $f(ax + b)$, the primitive is $\frac{1}{a}F(ax + b) + C$. Check by differentiating mentally.
4. Add the constant C: Never forget the arbitrary constant for indefinite integrals.

EXEMPLE **Algebraic and Trigonometric Combination**

Find the following indefinite integrals:

1. $\int \left(3x^2 - \frac{4}{x} + 5e^{2x}\right) dx$
2. $\int \frac{\sin 2x}{\cos^2 2x}\, dx$
3. $\int \frac{x^3 + 2x^2 - x + 1}{x^2}\, dx$

WORKED EXAMPLE 15 **Indefinite Integration: Worked Solutions**

(a) Apply linearity and standard forms term by term.

$$\int \left(3x^2 - \frac{4}{x} + 5e^{2x}\right) dx = 3\int x^2\, dx - 4\int \frac{1}{x}\, dx + 5\int e^{2x}\, dx$$

$$= 3\left(\frac{x^3}{3}\right) - 4\ln|x| + 5\left(\frac{e^{2x}}{2}\right) + C = \boxed{x^3 - 4\ln|x| + \frac{5}{2}e^{2x} + C}$$

(b) Recognise the derivative of $\sec 2x$ or use substitution $u = \cos 2x$. Method 1 (Inspection): $\frac{d}{dx}(\sec 2x) = 2\sec 2x \tan 2x = \frac{2\sin 2x}{\cos^2 2x}$. The integrand is $\frac{1}{2}$ of this derivative.

$$\int \frac{\sin 2x}{\cos^2 2x}\, dx = \frac{1}{2}\sec 2x + C$$

Method 2 (Substitution $u = \cos 2x$, $du = -2\sin 2x\, dx$):

$$\int \frac{\sin 2x}{\cos^2 2x}\, dx = \int \frac{-1/2\, du}{u^2} = -\frac{1}{2}\int u^{-2}\, du = -\frac{1}{2}\left(\frac{u^{-1}}{-1}\right) + C = \frac{1}{2u} + C = \boxed{\frac{1}{2}\sec 2x + C}$$

(c) The degree of numerator (≥ degree of denominator). Perform long division first.

$$\frac{x^3 + 2x^2 - x + 1}{x^2} = x + 2 - \frac{1}{x} + \frac{1}{x^2}$$

Now integrate term by term:

$$\int \left(x + 2 - x^{-1} + x^{-2}\right) dx = \frac{x^2}{2} + 2x - \ln|x| + \frac{x^{-1}}{-1} + C$$

$$= \boxed{\frac{x^2}{2} + 2x - \ln|x| - \frac{1}{x} + C}$$

WARNING **Common Mistakes**

- Forgetting the absolute value in $\ln|x|$ when integrating $1/x$.

- Forgetting the factor $1/a$ when integrating $f(ax+b)$ (e.g. $\int \cos 3x\,dx = \frac{1}{3}\sin 3x + C$, not $\sin 3x + C$).
- Omitting the constant C in indefinite integrals.
- Not simplifying the integrand before integrating (e.g. long division, trigonometric identities).

9 DEFINITE INTEGRALS AND THE FUNDAMENTAL THEOREM

DEFINITION **Definite Integral**

Let f be continuous on $[a,b]$. The **definite integral** of f from a to b is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x,$$

where $\Delta x = \frac{b-a}{n}$ and x_i^* is a sample point in the i-th subinterval. It represents the **signed area** between the curve $y = f(x)$ and the x-axis on $[a,b]$.

PROPERTY **Fundamental Theorem of Calculus (FTC)**

If f is continuous on $[a,b]$ and F is any primitive of f on $[a,b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a) = \Big[F(x)\Big]_a^b.$$

Proof (examinable): Define $A(x) = \int_a^x f(t)\,dt$. Show $A'(x) = f(x)$ using the Mean Value Theorem for integrals. Then $A(x) = F(x)+C$; evaluate at $x = a$ to find $C = -F(a)$; then $A(b) = F(b)-F(a)$.

PROPERTY **Properties of Definite Integrals**

For f, g continuous on $[a,b]$ and $c \in [a,b]$:

1. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
2. $\int_a^a f(x)\,dx = 0$

3. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
4. $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
5. $\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$

METHOD **Evaluating Definite Integrals: FTC**

Goal: Compute $\int_a^b f(x)\,dx = F(b) - F(a)$.
Steps:

1. Find the Primitive $F(x)$: Compute $\int f(x)\,dx$ (omit $+C$).
2. Apply Limits: Write $\Big[F(x)\Big]_a^b$. Substitute upper limit b, then subtract value at lower limit a.
3. Simplify: Combine logarithms ($\ln A - \ln B = \ln \frac{A}{B}$), rationalise denominators, evaluate trig functions at standard angles exactly.
4. Check Continuity: FTC requires f continuous on $[a, b]$. If f has a vertical asymptote inside $[a, b]$, the integral is improper (split at asymptote, take limits).

Reflex: Always write the square bracket notation $\Big[F(x)\Big]_a^b$ before substituting numbers to avoid sign errors.

WORKED EXAMPLE 16 **Definite Integrals: Exponential, Rational and Trigonometric**

Evaluate exactly:

1. $\displaystyle\int_0^1 \frac{e^x + e^{-x}}{2}\,dx$
2. $\displaystyle\int_1^2 \frac{2x+3}{x^2+3x}\,dx$
3. $\displaystyle\int_0^{\pi/4} \tan x\,dx$

SOLUTION

Detailed Solutions:
(a) Recognise the hyperbolic cosine: $\cosh x = \frac{e^x+e^{-x}}{2}$. Primitive: $\int \cosh x\,dx = \sinh x = \frac{e^x-e^{-x}}{2}$.

$$\int_0^1 \cosh x\,dx = \Big[\sinh x\Big]_0^1 = \sinh 1 - \sinh 0 = \frac{e^1 - e^{-1}}{2} - 0 = \boxed{\frac{e - e^{-1}}{2}}$$

Alternative without hyperbolic functions:

$$\int_0^1 \frac{e^x + e^{-x}}{2}\,dx = \frac{1}{2}\left[e^x - e^{-x}\right]_0^1 = \frac{1}{2}\left((e - e^{-1}) - (1 - 1)\right) = \frac{e - e^{-1}}{2}$$

(b) The numerator is the derivative of the denominator. Let $u = x^2 + 3x$, $du = (2x + 3)\,dx$. Limits: $x = 1 \Rightarrow u = 4$; $x = 2 \Rightarrow u = 10$.

$$\int_1^2 \frac{2x + 3}{x^2 + 3x}\,dx = \int_4^{10} \frac{du}{u} = \left[\ln|u|\right]_4^{10} = \ln 10 - \ln 4 = \ln\frac{10}{4} = \boxed{\ln\frac{5}{2}}$$

Directly: $\left[\ln|x^2 + 3x|\right]_1^2 = \ln|10| - \ln|4| = \ln\frac{5}{2}$.

(c) $\tan x = \frac{\sin x}{\cos x}$. Primitive is $-\ln|\cos x|$ (or $\ln|\sec x|$).

$$\int_0^{\pi/4} \tan x\,dx = \left[-\ln|\cos x|\right]_0^{\pi/4} = -\ln\left(\cos\frac{\pi}{4}\right) + \ln(\cos 0)$$

$$= -\ln\left(\frac{\sqrt{2}}{2}\right) + \ln(1) = -\ln(2^{-1/2}) = \frac{1}{2}\ln 2 = \boxed{\frac{1}{2}\ln 2}$$

EXPERIMENT **Area Interpretation**

Use graphing software (or sketch by hand) to plot $y = \frac{e^x + e^{-x}}{2}$ on $[0, 1]$. The area under the curve equals $\frac{e - e^{-1}}{2} \approx 1.175$. Compare with the rectangle of height 1 (area 1) and the trapezoid joining $(0, 1)$ and $(1, \cosh 1)$ (area ≈ 1.27). The integral lies between them, as expected for a convex function.

10 INTEGRATION OF RATIONAL FUNCTIONS

DEFINITION **Rational Function**

A **rational function** is a quotient $R(x) = \frac{P(x)}{Q(x)}$ where P, Q are polynomials. It is **proper** if $\deg P < \deg Q$, otherwise **improper**.

METHOD **Integration of Rational Functions: Partial Fractions**

Goal: Integrate $\int \frac{P(x)}{Q(x)}\,dx$.
Steps:

1. Check Degree: If $\deg P \geq \deg Q$, perform polynomial long division. Write $\frac{P}{Q} = S(x) + \frac{R(x)}{Q(x)}$ with $\deg R < \deg Q$.
2. Factorise Denominator $Q(x)$: Over $\mathbb{R}$ into linear factors $(ax + b)^m$ and irreducible quadratics $(ax^2 + bx + c)^n$ $(\Delta < 0)$.
3. Set up Partial Fractions Decomposition:
   - Linear factor $(px + q)^m$: $\frac{A_1}{px+q} + \frac{A_2}{(px+q)^2} + \cdots + \frac{A_m}{(px+q)^m}$
   - Quadratic factor $(ax^2 + bx + c)^n$: $\frac{B_1x+C_1}{ax^2+bx+c} + \cdots + \frac{B_nx+C_n}{(ax^2+bx+c)^n}$
4. Determine Constants: Use Heaviside cover-up for distinct linear factors, or equate coefficients / substitute convenient x-values for repeated/quadratic factors.
5. Integrate Each Term:

$$\int \frac{A}{(px+q)^k}\,dx = \begin{cases} \frac{A}{p}\ln|px+q| & k = 1 \\ \frac{A}{p(1-k)}(px+q)^{1-k} & k > 1 \end{cases}$$

For $\int \frac{Bx+C}{ax^2+bx+c}\,dx$: Split into $\frac{B}{2a}\frac{2ax+b}{ax^2+bx+c}$ + constant × $\frac{1}{ax^2+bx+c}$. The first part gives a logarithm; the second gives an arctan (complete the square).

WORKED EXAMPLE 17 **Rational Functions: Partial Fractions with Repeated and Quadratic Factors**

Evaluate $\int \frac{3x^3 + 2x^2 + 7x + 4}{(x^2 + 1)(x + 1)^2}\,dx$.

SOLUTION

Detailed Solution:
Step 1: Check Degree. Numerator degree 3, Denominator degree 4. Proper fraction. No long division needed.
Step 2: Factorise Denominator. $(x^2 + 1)$ (irreducible quadratic), $(x + 1)^2$ (repeated linear).
Step 3: Set up Decomposition.

$$\frac{3x^3 + 2x^2 + 7x + 4}{(x^2 + 1)(x + 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x + 1} + \frac{D}{(x + 1)^2}$$

Step 4: Find Constants A, B, C, D. Multiply by $(x^2 + 1)(x + 1)^2$:

$$3x^3 + 2x^2 + 7x + 4 = (Ax + B)(x + 1)^2 + C(x^2 + 1)(x + 1) + D(x^2 + 1)$$

Substitute $x = -1$ (Cover-up for D): LHS $= 3(-1) + 2(1) + 7(-1) + 4 = -3 + 2 - 7 + 4 = -4$. RHS $= D(1 + 1) = 2D \Rightarrow D = -2$.
Substitute $x = 0$: LHS $= 4$. RHS $= B(1) + C(1)(1) + D(1) = B + C - 2 \Rightarrow B + C = 6$. **(Eq 1)**
Substitute $x = 1$: LHS $= 3 + 2 + 7 + 4 = 16$. RHS $= (A + B)(4) + C(2)(2) + D(2) = 4A + 4B + 4C - 4$. $4A + 4B + 4C = 20 \Rightarrow A + B + C = 5$. **(Eq 2)**
Compare x^3 coefficients: LHS: 3. RHS: A (from $Ax \cdot x^2$) $+C$ (from $C \cdot x^2 \cdot x$) $= A + C$. $\Rightarrow A + C = 3$. (Eq 3)
Solve System: Eq 2 - Eq 1: $(A + B + C) - (B + C) = 5 - 6 \Rightarrow A = -1$. Eq 3: $-1 + C = 3 \Rightarrow C = 4$. Eq 1: $B + 4 = 6 \Rightarrow B = 2$.
Decomposition:

$$\frac{-x + 2}{x^2 + 1} + \frac{4}{x + 1} - \frac{2}{(x + 1)^2}$$

Step 5: Integrate.

$$\int \frac{-x + 2}{x^2 + 1}\,dx + \int \frac{4}{x + 1}\,dx - \int \frac{2}{(x + 1)^2}\,dx$$

Split first integral: $\int \frac{-x}{x^2+1}\,dx + \int \frac{2}{x^2+1}\,dx$.

$$\int \frac{-x}{x^2 + 1}\,dx = -\frac{1}{2}\ln(x^2 + 1) \quad \text{(sub } u = x^2 + 1\text{)}$$
$$\int \frac{2}{x^2 + 1}\,dx = 2\arctan x$$
$$\int \frac{4}{x + 1}\,dx = 4\ln|x + 1|$$
$$\int \frac{-2}{(x + 1)^2}\,dx = \frac{2}{x + 1} \quad \text{(power rule)}$$

Final Answer:

$$\boxed{-\frac{1}{2}\ln(x^2 + 1) + 2\arctan x + 4\ln|x + 1| + \frac{2}{x + 1} + C}$$

WARNING **Partial Fractions Traps**

- Always check degree first! Long division is mandatory for improper fractions.
- For irreducible quadratics, the numerator must be linear ($Bx + C$), not constant.
- When completing the square for $\int \frac{dx}{ax^2+bx+c}$, ensure the coefficient of x^2 is 1 before applying $\int \frac{du}{u^2+a^2} = \frac{1}{a}\arctan\frac{u}{a}$.
- Don't forget absolute values in logarithms of linear factors.

11 INTEGRATION BY SUBSTITUTION AND SYMMETRY

PROPERTY **Change of Variable Theorem (Substitution Rule)**

If $u = g(x)$ is differentiable on $[a, b]$ and f is continuous on the range of g, then

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

For indefinite integrals: $\int f(g(x))g'(x)\,dx = \int f(u)\,du$ with $u = g(x)$, then back-substitute.

METHOD **Integration by Substitution (Change of Variable)**

Goal: Simplify $\int f(g(x))g'(x)\,dx$ or $\int f(x)\,dx$ by letting $x = g(u)$ or $u = g(x)$.
Steps (Indefinite Integral):

1. Choose u: Identify an "inner function" $g(x)$ whose derivative $g'(x)$ appears (up to a constant factor) in the integrand. Common choices: u = radicand, u = denominator, u = trig argument.
2. Compute du: $du = g'(x)\,dx \implies dx = \frac{du}{g'(x)}$.
3. Substitute: Replace all x and dx with u and du. The new integral must be purely in u.
4. Integrate w.r.t u: Find $\int F(u)\,du = G(u) + C$.
5. Back-substitute: Replace u with $g(x)$ to express answer in terms of x.

Steps (Definite Integral $\int_a^b f(x)\,dx$):

1. Choose $u = g(x)$, find du.
2. Change Limits: $u_{\text{lower}} = g(a)$, $u_{\text{upper}} = g(b)$.
3. Integrate w.r.t u with new limits. No back-substitution needed.

Standard Trigonometric Substitutions (for $\sqrt{\cdot}$):

- $\sqrt{a^2 - x^2} \Rightarrow x = a\sin\theta$ ($dx = a\cos\theta\,d\theta$), $\theta \in [-\pi/2, \pi/2]$
- $\sqrt{a^2 + x^2} \Rightarrow x = a\tan\theta$ ($dx = a\sec^2\theta\,d\theta$), $\theta \in (-\pi/2, \pi/2)$
- $\sqrt{x^2 - a^2} \Rightarrow x = a\sec\theta$ ($dx = a\sec\theta\tan\theta\,d\theta$), $\theta \in [0, \pi/2) \cup (\pi/2, \pi]$

WORKED EXAMPLE 18 **Substitution: Algebraic and Trigonometric**

Evaluate:

1. $\int_0^1 \frac{x}{\sqrt{1+x^2}}\,dx$
2. $\int \frac{dx}{\sqrt{4-x^2}}$
3. $\int_0^{\pi/6} \frac{\cos x}{(1+\sin x)^2}\,dx$

SOLUTION

Detailed Solutions:

(a) Definite Integral: Algebraic Substitution. Let $u = 1+x^2 \Rightarrow du = 2x\,dx \Rightarrow x\,dx = \frac{1}{2}du$. Change limits: $x = 0 \Rightarrow u = 1$; $x = 1 \Rightarrow u = 2$.

$$\int_0^1 \frac{x}{\sqrt{1+x^2}}\,dx = \int_1^2 \frac{1}{\sqrt{u}} \cdot \frac{1}{2}\,du = \frac{1}{2}\int_1^2 u^{-1/2}\,du$$

$$= \frac{1}{2}\left[2u^{1/2}\right]_1^2 = \left[\sqrt{u}\right]_1^2 = \sqrt{2}-1 = \boxed{\sqrt{2}-1}$$

(b) Indefinite Integral: Trigonometric Substitution. Form $\sqrt{a^2-x^2}$ with $a = 2$. Use $x = 2\sin\theta$, $dx = 2\cos\theta\,d\theta, \theta \in [-\pi/2, \pi/2] \implies \cos\theta \geq 0$.

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2\theta} = \sqrt{4\cos^2\theta} = 2\cos\theta$$

$$\int \frac{dx}{\sqrt{4-x^2}} = \int \frac{2\cos\theta\,d\theta}{2\cos\theta} = \int d\theta = \theta + C$$

Back-substitute: $\sin\theta = \frac{x}{2} \Rightarrow \theta = \arcsin\frac{x}{2}$.

$$\boxed{\arcsin\frac{x}{2} + C}$$

(c) Definite Integral: Direct Substitution. Let $u = 1+\sin x \Rightarrow du = \cos x\,dx$. Change limits: $x = 0 \Rightarrow u = 1$; $x = \pi/6 \Rightarrow u = 1+\frac{1}{2} = \frac{3}{2}$.

$$\int_0^{\pi/6} \frac{\cos x}{(1+\sin x)^2}\,dx = \int_1^{3/2} \frac{du}{u^2} = \int_1^{3/2} u^{-2}\,du = \left[-u^{-1}\right]_1^{3/2}$$

$$= -\frac{1}{3/2} - (-1) = -\frac{2}{3} + 1 = \boxed{\frac{1}{3}}$$

METHOD **Definite Integrals of Symmetric Functions**

Goal: Exploit symmetry to evaluate $\int_{-a}^{a} f(x)\,dx$ quickly.
Properties (Proof Examinable): Let f be integrable on $[-a, a]$.

- Even Function ($f(-x) = f(x)$): $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$. Graph symmetric about y-axis.
- Odd Function ($f(-x) = -f(x)$): $\int_{-a}^{a} f(x)\,dx = 0$. Graph symmetric about origin; signed areas cancel.

Strategy:

1. Check if limits are symmetric $[-a, a]$.
2. Determine parity of $f(x)$: compute $f(-x)$.
3. If Even: Halve the interval, multiply by 2.
4. If Odd: Answer is 0 immediately.
5. If Neither: Split $f = f_{\text{even}} + f_{\text{odd}}$ (e.g. $x^3 + x^2 \to x^3$ (odd) + x^2 (even)). Integral of odd part is 0.

Warning: Only applies if f is defined and integrable on the whole interval $[-a, a]$. If there is a vertical asymptote at $x = 0$, the integral is improper; symmetry does not guarantee cancellation (e.g. $\int_{-1}^{1} \frac{1}{x}\,dx$ diverges).



Symmetry properties of definite integrals.

WORKED EXAMPLE 19 **Symmetry: Even, Odd, and Mixed Functions**

Evaluate:

1. $\int_{-2}^{2} (x^5 - 3x^3 + 2x)\,dx$

2. $\int_{-\pi/2}^{\pi/2} (x^2 \cos x + \sin x)\,dx$

3. $\int_{-1}^{1} \frac{x^4 + 2x^2 + 1}{x^2 + 1}\,dx$

SOLUTION

Detailed Solutions:

(a) $f(x) = x^5 - 3x^3 + 2x$. $f(-x) = (-x)^5 - 3(-x)^3 + 2(-x) = -x^5 + 3x^3 - 2x = -(x^5 - 3x^3 + 2x) = -f(x)$. f is odd. Limits $[-2, 2]$ are symmetric.

$$\boxed{\int_{-2}^{2} (x^5 - 3x^3 + 2x)\,dx = 0}$$

(b) $f(x) = x^2 \cos x + \sin x$. Check parity: $x^2 \cos x$: $(-x)^2 \cos(-x) = x^2 \cos x$ (Even × Even = Even). $\sin x$: $\sin(-x) = -\sin x$ (Odd). Sum = Even + Odd. Integral over $[-\pi/2, \pi/2]$:

$$\int_{-\pi/2}^{\pi/2} x^2 \cos x\,dx + \int_{-\pi/2}^{\pi/2} \sin x\,dx = 2\int_{0}^{\pi/2} x^2 \cos x\,dx + 0$$

Evaluate $I = \int_0^{\pi/2} x^2 \cos x\,dx$ by Parts (twice). Let $u = x^2, dv = \cos x\,dx \Rightarrow du = 2x\,dx, v = \sin x$. $I = \left[x^2 \sin x\right]_0^{\pi/2} - \int_0^{\pi/2} 2x \sin x\,dx = \frac{\pi^2}{4} - 2\int_0^{\pi/2} x \sin x\,dx$. For $J = \int x \sin x\,dx$: $u = x, dv = \sin x\,dx \Rightarrow du = dx, v = -\cos x$. $J = \left[-x\cos x\right]_0^{\pi/2} + \int_0^{\pi/2} \cos x\,dx = 0 + \left[\sin x\right]_0^{\pi/2} = 1$. $I = \frac{\pi^2}{4} - 2(1) = \frac{\pi^2}{4} - 2$. Total Integral $= 2I = \boxed{\frac{\pi^2}{2} - 4}$.

(c) $f(x) = \frac{x^4+2x^2+1}{x^2+1}$. Simplify first: Numerator $= (x^2 + 1)^2$. $f(x) = \frac{(x^2+1)^2}{x^2+1} = x^2 + 1$ (for all real x). $f(-x) = (-x)^2 + 1 = x^2 + 1 = f(x)$ (Even).

$$\int_{-1}^{1} (x^2 + 1)\,dx = 2\int_{0}^{1} (x^2 + 1)\,dx = 2\left[\frac{x^3}{3} + x\right]_0^1 = 2\left(\frac{1}{3} + 1\right) = \boxed{\frac{8}{3}}$$

DID YOU KNOW? **King's Property**

A powerful symmetry trick for integrals on $[0, a]$: $\int_0^a f(x)\,dx = \int_0^a f(a - x)\,dx$. This is a substitution $u = a - x$. Combined with the original integral, it often simplifies integrals involving x and trigonometric functions, as seen in the synthesis example $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\,dx$.

12 REDUCTION FORMULAE AND SYNTHESIS

DEFINITION **Reduction Formula**

A **reduction formula** expresses an integral I_n depending on an integer parameter n in terms of a similar integral with a lower index (I_{n-1} or I_{n-2}). It allows recursive evaluation of integrals with high powers.

METHOD **Deriving and Using Reduction Formulae**

Goal: Express $I_n = \int f_n(x)\,dx$ in terms of I_{n-1} or I_{n-2} to reduce the power/index n recursively.
Steps (Derivation – Examinable):

1. Define I_n clearly: e.g. $I_n = \int_0^{\pi/2} \sin^n x\,dx$ or $I_n = \int x^n e^x\,dx$.
2. Set up Integration by Parts: Choose u and dv to lower the power n.
   - For $\int \sin^n x\,dx$: Write $\sin^n x = \sin^{n-1} x \cdot \sin x$. Let $u = \sin^{n-1} x$, $dv = \sin x\,dx$.
   - For $\int x^n e^x\,dx$: Let $u = x^n$, $dv = e^x\,dx$.
   - For $\int \ln^n x\,dx$: Let $u = \ln^n x$, $dv = dx$.
3. Apply $\int u\,dv = uv - \int v\,du$: Obtain an expression involving I_n and I_{n-2} (or I_{n-1}).
4. Use Trig Identity (if needed): Replace $\cos^2 x = 1 - \sin^2 x$ or $\sec^2 x = 1 + \tan^2 x$ to get I_n on RHS.
5. Solve for I_n: Rearrange to get I_n = expression + kI_{n-2}.
6. State Base Cases: Compute I_0 and I_1 (or I_0, I_1 for definite) explicitly.

Steps (Application):

1. Write down the reduction formula.
2. Apply it repeatedly until reaching a base case (I_0 or I_1).
3. Substitute base values back up the chain.

WORKED EXAMPLE 20 **Reduction Formula: $\int \sin^n x\,dx$ and Application**

1. Derive the reduction formula for $I_n = \int \sin^n x\,dx$ $(n \geq 2)$.

2. Hence evaluate $\int_0^{\pi/2} \sin^6 x\,dx$.

SOLUTION

Detailed Solution:

(a) Derivation of Reduction Formula (Indefinite). Let $I_n = \int \sin^n x\,dx = \int \sin^{n-1} x \cdot \sin x\,dx$.

Integration by Parts: $u = \sin^{n-1} x \Rightarrow du = (n-1)\sin^{n-2} x \cos x\,dx$ $dv = \sin x\,dx \Rightarrow v = -\cos x$

$$I_n = -\sin^{n-1} x \cos x - \int (-\cos x)(n-1)\sin^{n-2} x \cos x\,dx$$

$$I_n = -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x \cos^2 x\,dx$$

Use $\cos^2 x = 1 - \sin^2 x$:

$$I_n = -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x(1-\sin^2 x)\,dx$$

$$I_n = -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x\,dx - (n-1)\int \sin^n x\,dx$$

$$I_n = -\sin^{n-1} x \cos x + (n-1)I_{n-2} - (n-1)I_n$$

Bring $(n-1)I_n$ to LHS: $I_n + (n-1)I_n = nI_n$.

$$nI_n = -\sin^{n-1} x \cos x + (n-1)I_{n-2}$$

$$\boxed{I_n = -\frac{1}{n}\sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}}$$

(b) Evaluate Definite Integral $J_n = \int_0^{\pi/2} \sin^n x\,dx$. For definite limits $[0, \pi/2]$, the boundary term vanishes: $\left[-\frac{1}{n}\sin^{n-1} x \cos x\right]_0^{\pi/2} = 0$. Reduction Formula: $\boxed{J_n = \frac{n-1}{n} J_{n-2}}$ for $n \geq 2$.

Base Cases: $J_0 = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$. $J_1 = \int_0^{\pi/2} \sin x\,dx = \left[-\cos x\right]_0^{\pi/2} = 1$.

Compute J_6 (Even n, reduces to J_0):

$$J_6 = \frac{5}{6} J_4$$

$$J_4 = \frac{3}{4} J_2$$

$$J_2 = \frac{1}{2} J_0 = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$$

Back-substitute: $J_4 = \frac{3}{4} \cdot \frac{\pi}{4} = \frac{3\pi}{16}$. $J_6 = \frac{5}{6} \cdot \frac{3\pi}{16} = \frac{15\pi}{96} = \boxed{\frac{5\pi}{32}}$.

KEY POINTS **Wallis's Formulae (Important for GCE)**

For $J_n = \int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx$:

$$J_n = \begin{cases} \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2} & n \text{ even} \\ \frac{(n-1)!!}{n!!} & n \text{ odd} \end{cases}$$

where $n!! = n(n-2)(n-4)\cdots$ (down to 2 or 1). These are derived directly from the reduction formula.

METHOD **Synthesis: Choosing the Right Technique**

Goal: Tackle integrals requiring a combination of methods (Substitution → Partial Fractions, Parts → Reduction, Symmetry → Substitution).
Decision Flowchart (Mental Checklist):

1. Simplify First: Long division? Trig identities ($\sin^2, \cos^2, \tan^2$)? Factorise?
2. Spot Standard Derivative: Is it $\int f'(x)/f(x)$, $\int f'(x)f(x)^n$, $\int f'(x)e^{f(x)}$?
3. Substitution ($u = g(x)$): Composite function? Radicand? Denominator?
4. Integration by Parts: Product of distinct types (Polynomial × Exp/Trig/Log)? $\int \ln x$, $\int \arcsin x$?
5. Partial Fractions: Proper rational function?
6. Trig Substitution: $\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 - a^2}$?
7. Symmetry: Limits $[-a, a]$ or $[0, a]$ with King's property?
8. Reduction: High power of trig/log/exp?

Verification: Always differentiate your answer mentally (or explicitly) to check.

WORKED EXAMPLE 21 **Synthesis Problem: GCE Style Mixed Integration**

Evaluate:

1. $\int_0^1 \frac{x^3}{\sqrt{1-x^2}}\,dx$
2. $\int \frac{x^2}{(x-1)^2(x+2)}\,dx$

3. $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\,dx$ (Hint: Use substitution $x \to \pi - x$)

SOLUTION

Detailed Solutions:

(a) $\int_0^1 \frac{x^3}{\sqrt{1-x^2}}\,dx$ Observation: Radicand $1-x^2$ suggests substitution $u = 1-x^2$ or trig sub $x = \sin\theta$. Method: Algebraic Substitution $u = 1-x^2$. $du = -2x\,dx \Rightarrow x\,dx = -\frac{1}{2}du$. $x^2 = 1-u$. Limits: $x = 0 \Rightarrow u = 1$; $x = 1 \Rightarrow u = 0$.

$$\int_0^1 \frac{x^3}{\sqrt{1-x^2}}\,dx = \int_0^1 \frac{x^2 \cdot x\,dx}{\sqrt{1-x^2}} = \int_1^0 \frac{(1-u)}{\sqrt{u}}\left(-\frac{1}{2}du\right)$$

$$= \frac{1}{2}\int_0^1 (1-u)u^{-1/2}\,du = \frac{1}{2}\int_0^1 (u^{-1/2} - u^{1/2})\,du$$

$$= \frac{1}{2}\left[2u^{1/2} - \frac{2}{3}u^{3/2}\right]_0^1 = \frac{1}{2}\left(2 - \frac{2}{3}\right) = \frac{1}{2}\cdot\frac{4}{3} = \boxed{\frac{2}{3}}$$

(b) $\int \frac{x^2}{(x-1)^2(x+2)}\,dx$ Observation: Proper rational function. Denominator factorised. Repeated linear factor. Partial Fractions:

$$\frac{x^2}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

$x^2 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$. $x = 1$: $1 = B(3) \Rightarrow B = 1/3$. $x = -2$: $4 = C(9) \Rightarrow C = 4/9$. Compare x^2 coeff: $1 = A + C \Rightarrow A = 1 - 4/9 = 5/9$. Integrate:

$$\int \left(\frac{5/9}{x-1} + \frac{1/3}{(x-1)^2} + \frac{4/9}{x+2}\right)dx$$

$$= \frac{5}{9}\ln|x-1| + \frac{1}{3}\frac{(x-1)^{-1}}{-1} + \frac{4}{9}\ln|x+2| + C$$

$$= \boxed{\frac{5}{9}\ln|x-1| - \frac{1}{3(x-1)} + \frac{4}{9}\ln|x+2| + C}$$

(c) $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,dx$ Observation: Limits $[0,\pi]$, numerator has x. Use King's Property: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. Let $I = \int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,dx$. Substitute $x \to \pi - x$: $I = \int_0^{\pi} \frac{(\pi-x)\sin(\pi-x)}{1+\cos^2(\pi-x)}\,dx = \int_0^{\pi} \frac{(\pi-x)\sin x}{1+(-\cos x)^2}\,dx = \int_0^{\pi} \frac{(\pi-x)\sin x}{1+\cos^2 x}\,dx$. Add the two expressions for I: $2I = \int_0^{\pi} \frac{x\sin x + (\pi-x)\sin x}{1+\cos^2 x}\,dx = \int_0^{\pi} \frac{\pi\sin x}{1+\cos^2 x}\,dx$. $I = \frac{\pi}{2}\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}\,dx$. Now substitute $u = \cos x$, $du = -\sin x\,dx$. Limits: $x = 0 \Rightarrow u = 1$; $x = \pi \Rightarrow u = -1$. $I = \frac{\pi}{2}\int_1^{-1} \frac{-du}{1+u^2} = \frac{\pi}{2}\int_{-1}^1 \frac{du}{1+u^2}$. Integrand $\frac{1}{1+u^2}$ is Even. Limits symmetric. $I = \frac{\pi}{2}\cdot 2\int_0^1 \frac{du}{1+u^2} = \pi\left[\arctan u\right]_0^1 = \pi\left(\frac{\pi}{4} - 0\right) = \boxed{\frac{\pi^2}{4}}$.

EXERCISE 6 Practice Exercises for Chapter FMA04

1. Find $\int \left(\frac{2}{x} - 3\sin 2x + 4e^{-x}\right) dx$.
2. Evaluate $\int_0^1 \frac{2x+1}{x^2+x+1}\, dx$.
3. Use partial fractions to find $\int \frac{5x-1}{(x-2)(x+3)}\, dx$.
4. Evaluate $\int_0^{\pi/2} \sin^2 x \cos^2 x\, dx$ using a reduction formula or trig identities.
5. Show that $\int_{-a}^{a} x^3 e^{x^2}\, dx = 0$ without integrating.
6. Derive the reduction formula for $I_n = \int x^n e^x\, dx$ and use it to find $\int x^3 e^x\, dx$.
7. Evaluate $\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\, dx$ using trigonometric substitution.
8. Find $\int \frac{dx}{x\sqrt{x^2-9}}$.
9. Use the substitution $u = \sqrt{x}$ to evaluate $\int_0^4 \frac{e^{\sqrt{x}}}{\sqrt{x}}\, dx$.
10. Evaluate $\int_0^{\pi} \frac{x}{1+\sin x}\, dx$ using King's property.

SOLUTION Exercise 1

$\int \left(\frac{2}{x} - 3\sin 2x + 4e^{-x}\right) dx = 2\ln|x| + \frac{3}{2}\cos 2x - 4e^{-x} + C.$

SOLUTION Exercise 2

Numerator is derivative of denominator: $\int_0^1 \frac{2x+1}{x^2+x+1}\, dx = \left[\ln(x^2 + x + 1)\right]_0^1 = \ln 3 - \ln 1 = \ln 3.$

SOLUTION Exercise 3

$\frac{5x-1}{(x-2)(x+3)} = \frac{9/5}{x-2} + \frac{16/5}{x+3}$. Integral $= \frac{9}{5}\ln|x-2| + \frac{16}{5}\ln|x+3| + C.$

SOLUTION Exercise 4

$\sin^2 x \cos^2 x = \frac{1}{4}\sin^2 2x = \frac{1}{8}(1-\cos 4x)$. $\int_0^{\pi/2} \frac{1}{8}(1-\cos 4x)\,dx = \frac{1}{8}\left[x - \frac{1}{4}\sin 4x\right]_0^{\pi/2} = \frac{\pi}{16}$.

SOLUTION **Exercise 5**

$f(x) = x^3 e^{x^2}$. $f(-x) = -x^3 e^{x^2} = -f(x)$ (odd). Limits symmetric $[-a, a]$. Integral = 0.

SOLUTION **Exercise 6**

$I_n = \int x^n e^x\,dx$. Parts: $u = x^n, dv = e^x dx \Rightarrow I_n = x^n e^x - nI_{n-1}$. $I_0 = e^x$. $I_1 = xe^x - e^x$. $I_2 = x^2e^x - 2(xe^x - e^x) = e^x(x^2 - 2x + 2)$. $I_3 = x^3e^x - 3e^x(x^2 - 2x + 2) = e^x(x^3 - 3x^2 + 6x - 6) + C$.

SOLUTION **Exercise 7**

$x = \sin\theta, dx = \cos\theta d\theta$. $\int_0^{\pi/2} \frac{\sin^2\theta}{\cos\theta}\cos\theta\,d\theta = \int_0^{\pi/2} \sin^2\theta\,d\theta = \int_0^{\pi/2} \frac{1-\cos 2\theta}{2}\,d\theta = \frac{\pi}{4}$.

SOLUTION **Exercise 8**

Standard form: $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\frac{|x|}{a} + C$. With $a = 3$: $\frac{1}{3}\operatorname{arcsec}\frac{|x|}{3} + C$.

SOLUTION **Exercise 9**

$u = \sqrt{x}, du = \frac{1}{2\sqrt{x}}dx \Rightarrow \frac{dx}{\sqrt{x}} = 2du$. Limits: $0 \to 0, 4 \to 2$. $\int_0^2 2e^u\,du = 2[e^u]_0^2 = 2(e^2 - 1)$.

SOLUTION **Exercise 10**

$I = \int_0^{\pi} \frac{x}{1+\sin x}\,dx$. King: $I = \int_0^{\pi} \frac{\pi - x}{1+\sin x}\,dx$. $2I = \pi\int_0^{\pi} \frac{dx}{1+\sin x}$. Multiply numerator/denominator by $1 - \sin x$: $\int_0^{\pi} \frac{1-\sin x}{\cos^2 x}\,dx = \int_0^{\pi}(\sec^2 x - \sec x\tan x)\,dx = [\tan x - \sec x]_0^{\pi}$. At π: $\tan\pi - \sec\pi = 0 - (-1) = 1$. At 0: $0 - 1 = -1$. Difference = 2. $2I = \pi \cdot 2 \Rightarrow I = \pi$.

13 EXERCISES

13.1 I check my knowledge

EXERCISE 7 **MCQ: Indefinite integrals**

Which of the following is the correct indefinite integral $\int \frac{1}{x^2+4}\,dx$?

1. $\frac{1}{2}\arctan\left(\frac{x}{2}\right)+C$
2. $\frac{1}{2}\ln(x^2+4)+C$
3. $\arctan(2x)+C$
4. $\frac{1}{2}\arctan(2x)+C$

EXERCISE 8 **True/False with justification**

Determine whether each statement is true or false. Justify your answer.

1. $\int_{-a}^{a} f(x)\,dx = 0$ for every function f and every $a > 0$.
2. If f is continuous on $[a, b]$, then $\int_a^b f(x)\,dx = F(b) - F(a)$ where F is any primitive of f.
3. The substitution $u = x^2$ transforms $\int x\cos(x^2)\,dx$ into $\frac{1}{2}\int \cos u\,du$.
4. A reduction formula always expresses I_n in terms of I_{n-1}.

EXERCISE 9 **Definitions and direct calculations**

1. Write the definition of the definite integral $\int_a^b f(x)\,dx$ as a limit of Riemann sums.
2. State the Fundamental Theorem of Calculus (Part 1 and Part 2).
3. Evaluate directly: $\int_0^1 (3x^2 - 2x + 5)\,dx$.
4. Evaluate directly: $\int \frac{2x}{x^2+1}\,dx$.

EXERCISE 10 **Quick integration practice**

Find the following indefinite integrals:

1. $\int \frac{dx}{x^2 - 9}$
2. $\int \frac{x}{\sqrt{1 - x^2}}\,dx$
3. $\int xe^{x^2}\,dx$
4. $\int \sin^2 x\,dx$ (use a trigonometric identity)

13.2 I practise

EXERCISE 11 **Integration of rational functions by partial fractions**

1. Express $\frac{5x - 4}{(x - 1)(x + 2)}$ in partial fractions.
2. Hence evaluate $\int_2^3 \frac{5x - 4}{(x - 1)(x + 2)}\,dx$, giving your answer in the form $\ln k$ where k is a rational number.

EXERCISE 12 **Integration by substitution**

1. Using the substitution $u = \sin x$, evaluate $\int_0^{\pi/2} \sin^3 x \cos x\,dx$.
2. Evaluate $\int_0^{\pi/2} \cos^3 x\,dx$ by writing $\cos^3 x = \cos x(1 - \sin^2 x)$ and using an appropriate substitution.

EXERCISE 13 **Symmetry and definite integrals**

Show that $\int_{-1}^{1} \frac{x^3 + x^2}{1 + x^2}\,dx$ reduces to a rational-function integral, and evaluate it using parity and the integral of $\frac{1}{1+x^2}$.

EXERCISE 14 **Algebraic division and integration**

Find $\int \frac{x^3 + 2x}{x^2 + 1}\,dx$ by first performing algebraic division.

13.3 I prepare for the GCE

EXERCISE 15 **GCE-style: Logarithmic integration and area**

(a) Show that $\frac{d}{dx}\left[x \ln x - x\right] = \ln x$.

(b) Hence find $\int_1^e \ln x\,dx$.

(c) Deduce the area between the curve $y = \ln x$ and the x-axis on the interval $\left[\frac{1}{e}, e\right]$.

(Total: 8 marks)

EXERCISE 16 **GCE-style: Reduction formula**

Given $I_n = \int_0^1 x^n e^x\,dx$ for $n \geq 0$.

(a) Prove that $I_n = e - nI_{n-1}$ for $n \geq 1$.

(b) Find the exact value of I_3.

(Total: 9 marks)

EXERCISE 17 **GCE-style: Mixed problem — velocity, substitution, mean value**

The velocity of a vehicle on the Yaoundé–Douala highway is modelled by $v(t) = te^{-t^2}$ m/s, where t is the time in seconds.

(a) Use the substitution $u = t^2$ to find the distance covered between $t = 0$ and $t = 2$.

(b) Find the mean velocity of the vehicle over the interval $[0, 2]$.

(Total: 7 marks)

14 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: Indefinite integrals**

The correct answer is (a): $\int \frac{1}{x^2+4}\,dx = \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$.

Justification. We use the standard result $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$ (valid for $a > 0$). Here $a^2 = 4$, so $a = 2$, giving

$$\int \frac{1}{x^2+4}\,dx = \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C.$$

Check by differentiation (chain rule, with $\frac{d}{dx}\arctan u = \frac{u'}{1+u^2}$):

$$\frac{d}{dx}\left[\frac{1}{2}\arctan\left(\frac{x}{2}\right)\right] = \frac{1}{2}\cdot\frac{1}{1+\left(\frac{x}{2}\right)^2}\cdot\frac{1}{2} = \frac{1}{4}\cdot\frac{1}{1+\frac{x^2}{4}} = \frac{1}{4+x^2}. \quad \checkmark$$

Option (b) would be correct for $\int \frac{2x}{x^2+4}\,dx = \ln(x^2+4) + C$ (logarithmic form, numerator equal to the derivative of the denominator). Options (c) and (d) use the wrong argument inside arctan.

SOLUTION **Exercise 2 — True/False with justification**

(a) FALSE. The result $\int_{-a}^{a} f(x)\,dx = 0$ holds only when f is odd. Counter-example: take $f(x) = 1$ (an even function). Then

$$\int_{-a}^{a} 1\,dx = [x]_{-a}^{a} = a - (-a) = 2a \neq 0.$$

(b) TRUE. This is precisely the Fundamental Theorem of Calculus (Part 2): if f is continuous on $[a,b]$ and F is any primitive of f on $[a,b]$, then $\int_a^b f(x)\,dx = F(b) - F(a)$. The choice of primitive does not matter, since two primitives differ by a constant C which cancels: $(F(b)+C) - (F(a)+C) = F(b) - F(a)$.

(c) TRUE. Let $u = x^2$. Then $du = 2x\,dx$, so $x\,dx = \frac{1}{2}\,du$. Hence

$$\int x\cos(x^2)\,dx = \int \cos u \cdot \frac{1}{2}\,du = \frac{1}{2}\int \cos u\,du = \frac{1}{2}\sin u + C = \frac{1}{2}\sin(x^2) + C.$$

Check: $\frac{d}{dx}\left[\frac{1}{2}\sin(x^2)\right] = \frac{1}{2}\cos(x^2)\cdot 2x = x\cos(x^2)$. $\checkmark$

(d) FALSE. A reduction formula expresses I_n in terms of an integral of the same family with a lower index, but not necessarily I_{n-1}. For example, for $I_n = \int_0^{\pi/2} \sin^n x\,dx$, integration by parts gives $I_n = \frac{n-1}{n} I_{n-2}$, which relates I_n to I_{n-2}, not I_{n-1}.

SOLUTION **Exercise 3 — Definitions and direct calculations**

(a) Definition as a limit of Riemann sums. Let f be defined on $[a, b]$. Divide $[a, b]$ into n subintervals of equal width $\Delta x = \frac{b-a}{n}$ with subdivision points $x_k = a + k\Delta x$, and choose a sample point x_k^* in each subinterval $[x_{k-1}, x_k]$. Then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\,\Delta x,$$

provided this limit exists and is independent of the choice of the sample points. Geometrically, each term $f(x_k^*)\Delta x$ is the area of a thin rectangle, and the limit gives the exact area under the curve.

(b) Fundamental Theorem of Calculus.

- Part 1: If f is continuous on $[a, b]$, then the function G defined by $G(x) = \int_a^x f(t)\,dt$ is differentiable on $[a, b]$ and $G'(x) = f(x)$. (Integration and differentiation are inverse operations.)
- Part 2: If f is continuous on $[a, b]$ and F is any primitive of f, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

(c) A primitive of $3x^2 - 2x + 5$ is $x^3 - x^2 + 5x$ (integrate term by term using $\int x^n\,dx = \frac{x^{n+1}}{n+1}$). Hence

$$\int_0^1 (3x^2 - 2x + 5)\,dx = \Big[x^3 - x^2 + 5x\Big]_0^1 = (1 - 1 + 5) - 0 = \boxed{5}.$$

(d) The numerator $2x$ is exactly the derivative of the denominator $x^2 + 1$, so this is the logarithmic form $\int \frac{u'}{u}\,dx = \ln|u| + C$:

$$\int \frac{2x}{x^2 + 1}\,dx = \ln(x^2 + 1) + C,$$

(the absolute value is unnecessary since $x^2 + 1 > 0$ for all x).

SOLUTION **Exercise 4 — Quick integration practice**

(a) $\int \frac{dx}{x^2 - 9}$. Using partial fractions: $\frac{1}{x^2 - 9} = \frac{1}{(x-3)(x+3)} = \frac{1/6}{x-3} - \frac{1/6}{x+3}$. Hence

$$\int \frac{dx}{x^2 - 9} = \frac{1}{6}\ln|x - 3| - \frac{1}{6}\ln|x + 3| + C = \frac{1}{6}\ln\left|\frac{x-3}{x+3}\right| + C.$$

(b) $\int \frac{x}{\sqrt{1 - x^2}}\,dx$. Let $u = 1 - x^2$, so $du = -2x\,dx$ and $x\,dx = -\frac{1}{2}du$:

$$\int \frac{x}{\sqrt{1 - x^2}}\,dx = -\frac{1}{2}\int u^{-1/2}\,du = -\frac{1}{2}\cdot 2u^{1/2} + C = -\sqrt{1 - x^2} + C.$$

Check: $\frac{d}{dx}\left[-\sqrt{1-x^2}\right] = -\frac{-2x}{2\sqrt{1-x^2}} = \frac{x}{\sqrt{1-x^2}}$. ✓

(c) $\int xe^{x^2}\,dx$. Let $u = x^2$, so $du = 2x\,dx$:

$$\int xe^{x^2}\,dx = \frac{1}{2}\int e^u\,du = \frac{1}{2}e^u + C = \frac{1}{2}e^{x^2} + C.$$

(d) $\int \sin^2 x\,dx$. Using the identity $\sin^2 x = \frac{1-\cos 2x}{2}$:

$$\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C = \frac{1}{2}(x - \sin x\cos x) + C,$$

since $\sin 2x = 2\sin x\cos x$.

SOLUTION **Exercise 5 — Integration of rational functions by partial fractions**

(a) We seek constants A and B such that

$$\frac{5x-4}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}.$$

Multiplying through by $(x-1)(x+2)$: $5x - 4 = A(x+2) + B(x-1)$.

- Setting $x = 1$: $5 - 4 = 3A$, so $A = \frac{1}{3}$.
- Setting $x = -2$: $-10 - 4 = -3B$, so $B = \frac{14}{3}$.

Therefore

$$\frac{5x-4}{(x-1)(x+2)} = \frac{1}{3(x-1)} + \frac{14}{3(x+2)}.$$

Check: $\frac{1}{3}(x+2) + \frac{14}{3}(x-1) = \frac{x+2+14x-14}{3} = \frac{15x-12}{3} = 5x - 4$. ✓

(b) Using the decomposition:

$$\begin{aligned}\int_2^3 \frac{5x-4}{(x-1)(x+2)}\,dx &= \frac{1}{3}\Big[\ln|x-1|\Big]_2^3 + \frac{14}{3}\Big[\ln|x+2|\Big]_2^3 \\ &= \frac{1}{3}(\ln 2 - \ln 1) + \frac{14}{3}(\ln 5 - \ln 4) \\ &= \frac{1}{3}\ln 2 + \frac{14}{3}\ln\frac{5}{4}.\end{aligned}$$

To write this as $\ln k$, combine the logarithms using $a\ln b = \ln(b^a)$ and $\ln p + \ln q = \ln(pq)$:

$$\frac{1}{3}\ln 2 + \frac{14}{3}\ln\frac{5}{4} = \frac{1}{3}\ln\left(2\cdot\left(\frac{5}{4}\right)^{14}\right) = \ln\left[2^{1/3}\left(\frac{5}{4}\right)^{14/3}\right].$$

Hence

$$k = 2^{1/3}\left(\frac{5}{4}\right)^{14/3} = \left(\frac{2\cdot 5^{14}}{4^{14}}\right)^{1/3} = \left(\frac{5^{14}}{2^{27}}\right)^{1/3} = \frac{5^{14/3}}{2^9} = \frac{5^4\cdot 5^{2/3}}{512}.$$

Numerical check: $\frac{1}{3}\ln 2 \approx 0.231$ and $\frac{14}{3}\ln\frac{5}{4} \approx 4.667 \times 0.2231 \approx 1.041$, so the integral is approximately 1.272, and indeed $k \approx \frac{625\times 2.924}{512} \approx 3.57$ with $\ln 3.57 \approx 1.272$. ✓

SOLUTION **Exercise 6 — Integration by substitution**

(a) Let $u = \sin x$, so $du = \cos x\,dx$. Change the limits: when $x = 0$, $u = 0$; when $x = \frac{\pi}{2}$, $u = 1$. Then

$$\int_0^{\pi/2} \sin^3 x \cos x\,dx = \int_0^1 u^3\,du = \left[\frac{u^4}{4}\right]_0^1 = \frac{1}{4}.$$

(b) Write $\cos^3 x = \cos x\,(1 - \sin^2 x)$. Let $u = \sin x$, so $du = \cos x\,dx$, with the same change of limits as above:

$$\int_0^{\pi/2} \cos^3 x\,dx = \int_0^{\pi/2} (1 - \sin^2 x)\cos x\,dx = \int_0^1 (1 - u^2)\,du = \left[u - \frac{u^3}{3}\right]_0^1 = 1 - \frac{1}{3} = \frac{2}{3}.$$

Remark: this agrees with the reduction formula $I_3 = \frac{2}{3}I_1 = \frac{2}{3} \times 1$ for $I_n = \int_0^{\pi/2} \cos^n x\,dx$, since $I_1 = \int_0^{\pi/2} \cos x\,dx = 1$.

SOLUTION **Exercise 7 — Symmetry and definite integrals**

Split the integrand:

$$\frac{x^3 + x^2}{1 + x^2} = \underbrace{\frac{x^3}{1 + x^2}}_{\text{odd}} + \underbrace{\frac{x^2}{1 + x^2}}_{\text{even}}.$$

Indeed, if $g(x) = \frac{x^3}{1+x^2}$, then $g(-x) = \frac{-x^3}{1+x^2} = -g(x)$ (odd), and if $h(x) = \frac{x^2}{1+x^2}$, then $h(-x) = h(x)$ (even).
Since the interval $[-1, 1]$ is symmetric about 0:

$$\int_{-1}^1 \frac{x^3}{1 + x^2}\,dx = 0 \quad \text{(odd function)}, \qquad \int_{-1}^1 \frac{x^2}{1 + x^2}\,dx = 2\int_0^1 \frac{x^2}{1 + x^2}\,dx \quad \text{(even function)}.$$

For the remaining rational-function integral, write $\frac{x^2}{1 + x^2} = \frac{(1 + x^2) - 1}{1 + x^2} = 1 - \frac{1}{1 + x^2}$:

$$2\int_0^1 \left(1 - \frac{1}{1 + x^2}\right)dx = 2\Big[x - \arctan x\Big]_0^1 = 2\left(1 - \frac{\pi}{4}\right) = 2 - \frac{\pi}{2}.$$

Therefore

$$\int_{-1}^1 \frac{x^3 + x^2}{1 + x^2}\,dx = 2 - \frac{\pi}{2}.$$

SOLUTION **Exercise 8 — Algebraic division and integration**

Since the degree of the numerator (3) exceeds that of the denominator (2), perform al-

gebraic division:

$$x^3 + 2x = x(x^2 + 1) + x,$$

so

$$\frac{x^3 + 2x}{x^2 + 1} = x + \frac{x}{x^2 + 1}.$$

Now integrate term by term. The first term gives $\frac{x^2}{2}$. For the second, the numerator is (half) the derivative of the denominator:

$$\int \frac{x}{x^2 + 1}\,dx = \frac{1}{2}\int \frac{2x}{x^2 + 1}\,dx = \frac{1}{2}\ln(x^2 + 1) + C.$$

Therefore

$$\int \frac{x^3 + 2x}{x^2 + 1}\,dx = \frac{x^2}{2} + \frac{1}{2}\ln(x^2 + 1) + C.$$

Check by differentiation: $\frac{d}{dx}\left[\frac{x^2}{2} + \frac{1}{2}\ln(x^2 + 1)\right] = x + \frac{x}{x^2+1} = \frac{x(x^2+1)+x}{x^2+1} = \frac{x^3+2x}{x^2+1}$. ✓

SOLUTION **Exercise 9 — GCE-style: Logarithmic integration and area**

(a) (2 marks) Differentiate using the product rule on $x \ln x$:

$$\frac{d}{dx}[x\ln x - x] = \left(1 \cdot \ln x + x \cdot \frac{1}{x}\right) - 1 = \ln x + 1 - 1 = \ln x. \qquad \blacksquare$$

(b) (2 marks) Part (a) shows that $x \ln x - x$ is a primitive of $\ln x$. Hence

$$\int_1^e \ln x\,dx = \Big[x\ln x - x\Big]_1^e = (e\ln e - e) - (1 \cdot \ln 1 - 1) = (e - e) - (0 - 1) = 1.$$

(c) (4 marks) On $\left[\frac{1}{e}, 1\right]$, $\ln x \le 0$, so the curve lies below the x-axis; on $[1, e]$, $\ln x \ge 0$, so the curve lies above. The area requires the integral of $|\ln x|$, so we split at $x = 1$:

$$A = -\int_{1/e}^{1} \ln x\,dx + \int_1^e \ln x\,dx.$$

Compute the first integral:

$$\int_{1/e}^{1} \ln x\,dx = \Big[x\ln x - x\Big]_{1/e}^{1} = (0 - 1) - \left(\frac{1}{e}\ln\frac{1}{e} - \frac{1}{e}\right) = -1 - \left(-\frac{1}{e} - \frac{1}{e}\right) = -1 + \frac{2}{e}.$$

From (b), $\int_1^e \ln x\,dx = 1$. Therefore

$$A = -\left(-1 + \frac{2}{e}\right) + 1 = 2 - \frac{2}{e} = 2\left(1 - \frac{1}{e}\right) \text{ square units.}$$

Examiner's expectations: In (a), correct use of the product rule (1 mark) and clean simplification (1 mark). In (b), correct evaluation at both bounds with $\ln e = 1$ and $\ln 1 = 0$. In (c), the key marking point is recognising the change of sign of $\ln x$ at $x = 1$ and splitting the integral (2 marks); a candidate who integrates straight from $\frac{1}{e}$ to e obtains $\frac{2}{e}$ instead of the area and loses these marks.

SOLUTION **Exercise 10 — GCE-style: Reduction formula**

(a) (4 marks) $I_n = \int_0^1 x^n e^x \, dx$. Integrate by parts with

$$u = x^n \quad \Rightarrow \quad du = nx^{n-1}\,dx; \qquad dv = e^x\,dx \quad \Rightarrow \quad v = e^x.$$

Then

$$\begin{aligned} I_n &= \left[x^n e^x\right]_0^1 - \int_0^1 nx^{n-1}e^x\,dx \\ &= \left(1^n \cdot e - 0^n \cdot e^0\right) - n\int_0^1 x^{n-1}e^x\,dx \\ &= e - nI_{n-1}, \qquad n \geq 1. \quad \blacksquare \end{aligned}$$

(Note: $0^n = 0$ since $n \geq 1$; this is why the condition $n \geq 1$ is needed.)
(b) (5 marks) First compute I_0 directly:

$$I_0 = \int_0^1 e^x\,dx = \left[e^x\right]_0^1 = e - 1.$$

Now apply the reduction formula repeatedly:

$$\begin{aligned} I_1 &= e - 1 \cdot I_0 = e - (e-1) = 1, \\ I_2 &= e - 2I_1 = e - 2, \\ I_3 &= e - 3I_2 = e - 3(e-2) = e - 3e + 6 = 6 - 2e. \end{aligned}$$

Therefore $\boxed{I_3 = 6 - 2e}$.
Sanity check: $6 - 2e \approx 6 - 5.437 = 0.563 > 0$, consistent with the integral of a positive function; also $I_3 < I_0 \approx 1.718$, as expected since $x^3 \leq 1$ on $[0,1]$.
Examiner's expectations: In (a), correct choice of u and dv (1 mark), correct by-parts formula (1 mark), correct evaluation of the boundary term including $0^n = 0$ (1 mark), and recognising the remaining integral as I_{n-1} (1 mark). In (b), computing I_0 (1 mark) and three correct applications of the recurrence (3 marks), with the exact simplified answer (1 mark).

SOLUTION **Exercise 11 — GCE-style: Mixed problem — velocity, substitution, mean value**

(a) (4 marks) The distance covered between $t = 0$ and $t = 2$ is

$$s = \int_0^2 v(t)\,dt = \int_0^2 te^{-t^2}\,dt.$$

Let $u = t^2$, so $du = 2t\,dt$, i.e. $t\,dt = \frac{1}{2}du$. Change the limits: when $t = 0$, $u = 0$; when $t = 2$,

$u = 4$. Then

$$s = \frac{1}{2}\int_0^4 e^{-u}\,du = \frac{1}{2}\Big[-e^{-u}\Big]_0^4 = \frac{1}{2}\left(-e^{-4}+1\right) = \frac{1-e^{-4}}{2}.$$

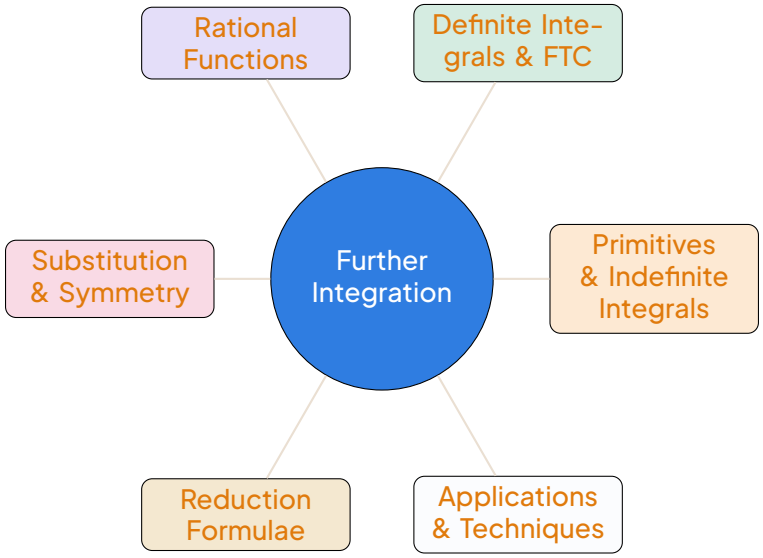
So the distance covered is $s = \dfrac{1-e^{-4}}{2} \approx 0.491\,\text{m}$.

(b) (3 marks) The mean velocity over $[0,2]$ is

$$\bar{v} = \frac{1}{2-0}\int_0^2 v(t)\,dt = \frac{1}{2}\cdot\frac{1-e^{-4}}{2} = \frac{1-e^{-4}}{4} \approx 0.245\,\text{m s}^{-1}.$$

Examiner's expectations: In (a), knowing that distance is the integral of velocity (1 mark), a correct substitution with $du = 2t\,dt$ (1 mark), correct change of limits (1 mark), and the exact final answer (1 mark). In (b), correct statement of the mean-value formula $\bar{v} = \frac{1}{b-a}\int_a^b v\,dt$ (1 mark) and correct use of the result of (a) (2 marks). A common error is forgetting to change the limits and mixing variables — penalised in the GCE.

15 SUMMARY SHEET



Mind map of the Further Integration chapter (FMA04)

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key Definitions

- **Primitive**: A function F such that $F'(x) = f(x)$ on an interval. Denoted $\int f(x)\,dx = F(x) + C$.
- **Indefinite Integral**: The family of all primitives of f, written $\int f(x)\,dx$.
- **Definite Integral**: $\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$, represents signed area.
- **Fundamental Theorem of Calculus**: If F is a primitive of f on $[a, b]$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.
- **Rational Function**: $R(x) = \frac{P(x)}{Q(x)}$ with P, Q polynomials, $\deg P < \deg Q$ (proper) or use long division.
- **Reduction Formula**: A recurrence I_n = expression involving I_{n-1} or I_{n-2}.

Laws, Formulas, Theorems to Know

- Linearity: $\int (af + bg) = a\int f + b\int g$.
- Power rule: $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$ $(n \neq -1)$; $\int \frac{dx}{x} = \ln|x| + C$.
- Exponential/Log: $\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$; $\int a^x\,dx = \frac{a^x}{\ln a} + C$.
- Trigonometric: $\int \sin ax\,dx = -\frac{1}{a}\cos ax + C$, $\int \cos ax\,dx = \frac{1}{a}\sin ax + C$, $\int \sec^2 ax\,dx =$

$\frac{1}{a}\tan ax + C$, etc.

- Hyperbolic: $\int \sinh ax\,dx = \frac{1}{a}\cosh ax + C$, $\int \cosh ax\,dx = \frac{1}{a}\sinh ax + C$.
- Inverse trigonometric: $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C$, $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a} + C$.
- Partial fractions: Decompose proper rational function into sum of simpler fractions (linear/quadratic factors).
- Substitution: $\int f(g(x))g'(x)\,dx = \int f(u)\,du$ with $u = g(x)$.
- Symmetry: If f even, $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$; if f odd, $\int_{-a}^{a} f(x)\,dx = 0$.
- Reduction formulae (examples): $I_n = \int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}I_{n-2}$, $J_n = \int \cos^n x\,dx = \frac{1}{n}\cos^{n-1}x\sin x + \frac{n-1}{n}J_{n-2}$, $K_n = \int x^n e^x\,dx = x^n e^x - nK_{n-1}$.

Methods (Step-by-Step)

1. Identify the type: polynomial, rational, trigonometric, exponential, composite.
2. Simplify: long division for improper rational functions; use trigonometric identities.
3. Choose technique:
   - Direct integration (standard forms).
   - Substitution $u = g(x)$ when $g'(x)$ appears.
   - Partial fractions for rational functions.
   - Integration by parts (if product of functions).
   - Reduction formulae for powers of trig/exponential.
   - Symmetry for definite integrals over symmetric intervals.
4. Execute: perform algebraic manipulation, integrate term by term.
5. Back-substitute: return to original variable.
6. Check: differentiate result; for definite integrals, apply limits.

Common Mistakes (Traps)

- Forgetting the constant C in indefinite integrals.
- Incorrect partial fraction decomposition (missing terms for repeated/quadratic factors).
- Misapplying substitution: not changing limits for definite integrals or forgetting $du = g'(x)dx$.
- Sign errors in reduction formulae (especially with $\sin^n$ and $\cos^n$).
- Assuming $\int f(x)g(x)\,dx = \int f(x)\,dx \int g(x)\,dx$ (false).

- Ignoring absolute value in $\ln|x|$ when integrating $1/x$.
- Misusing symmetry: function must be even/odd on the whole interval $[-a, a]$.

GCE Examination Tips

- Always show the substitution step clearly: state u, du, change limits if definite.
- For partial fractions, write the form before solving for constants.
- When using reduction formulae, state the formula and the base cases (I_0, I_1).
- For definite integrals, write $F(b) - F(a)$ explicitly.
- Simplify final answers: combine logarithms, rationalize denominators.
- Check if the integral is improper (infinite limits or discontinuities) – mention convergence.
- Practice standard integrals: $\int \frac{dx}{x^2+a^2}$, $\int \frac{dx}{\sqrt{a^2-x^2}}$, $\int \frac{dx}{x^2-a^2}$.
- Time management: do straightforward integrals first; leave reduction formulae for later if time permits.

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